

UFMF8T-60-M

Masters Group Capstone Project

Individual Technical Report: 1/4 Scale Proof-of-Concept Model Turntable Mounting System & LabVIEW Control

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1 Summary

1.1 System Overview

This report describes the design of a 1/4 scale proof-of-concept (PoC) turntable mounting system for the UWE large subsonic wind tunnel, as part of the wider tunnel upgrade project. The turntable enables accurate positioning of a test model within the test section, with position control implemented on LabVIEW. The work addresses Objective 2 (Manufacturing) from the project brief. The system is designed as a standalone PoC with well-defined interfaces, wiring documentation, and a LabVIEW SubVI connector to allow for future integration.

1.2 Key Design Challenges

Positional accuracy was the main engineering challenge. A target resolution of 1° and positional accuracy of $\pm 0.3^\circ$ was required to ensure quality aerodynamic data. Achieving this with a low-cost, timber-based PoC structure, without the precision of machined components, required careful selection of the drive mechanism and actuation system. An open-loop stepper was identified from the literature as insufficient under high aerodynamic loading, making closed-loop control the sensible design choice.

Control system integration was an additional challenge. Configuring the CL42T driver to produce the correct pulses per revolution to achieve the target resolution with the 3:1 reduction and coding the ESP32 controller to generate the pulses correctly, posed a challenge, due to limited experience with closed-loop stepper systems and C++ programming. This was resolved through reference to the CL42T manual, iterative testing, and use of online resources.

Wireless communication was required to allow integration without routing control wiring to the tunnel control room. However, this posed a problem firstly with choosing the most suitable protocol to send the motor control commands wirelessly and then configuring the protocol chosen (ESP-NOW) successfully. This was overcome through a comparative review of wireless communication protocols and learning how to configure the communication with the help of online resources.

1.3 Outcomes

The designed system successfully achieves wireless positional control on the turntable shaft through a LabVIEW control interface, with two operating modes: manual angle control and oscillation mode. The system achieves the target resolution of 1° - the resolution was limited to 1° increments in the LabVIEW control interface, as a finer resolution was not required by the system specification, however the drive system is theoretically capable of 0.075° per step. Positional accuracy against the $\pm 0.3^\circ$ target set in the specification could not be experimentally validated within the project timeline due to the encoder mount not being completed in time – this is recommended for future work in Section 8.4. The connection between the disc and drive shaft, through the 3D printed shaft housing, was found to be insufficient during final testing, resulting in the shaft rotating without transmitting motion to the disc – this is identified as a limitation in Section 8.3 with a proposed solution.

2 Overview

2.1 System Description

The automated turntable system will allow angular rotation along the vertical axis of models installed in UWE's large subsonic wind tunnel's small test section with a resolution of 1° , and an accuracy of $\pm 0.3^\circ$. The system will allow controlled rotation of a model's yaw or AoA depending on how the model is mounted relative to the oncoming flow. The system will offer three different modes: manual angle control and oscillation mode – the final mode offering function that would not be available with the previous manual model orientation method.

The quarter scale PoC represents a scaled down version of the two dimensional constraints defined – the desired turntable disc diameter of 1005 mm and height available underneath the test section of 733 mm. This gives the PoC a disc diameter of 280 mm and a height of 190 mm, approximately a quarter of both constraints. The frame of the turntable is a triangular prism shape, with the disc supported on the frame with a lazy Susan bearing and rotated by a central

shaft; the central shaft is supported through the frame by two 608ZZ bearings. The shaft itself is rotated through a GT2 belt drive system, with a 60-tooth pulley connected to the central shaft, connected to a 20-tooth pulley, on the motor shaft giving a 3:1 reduction. The motor itself is a closed-loop NEMA 17 stepper motor driven by a CL42T closed-loop driver.

The system is controlled in a LabVIEW control environment on an external PC in a control room, the control commands are sent to an ESP32 microcontroller transmitter that then transmits the command wirelessly, through ESP-NOW protocol to an ESP32 receiver, which sends the command to the motor driver which then drives the motor, with position verified and corrected in real time by the driver using the motor's internal encoder.

3 Literature Review

3.1 Introduction

Small errors in model orientation during wind tunnel testing transfers directly into measurement error, a 1° change in angle of attack changes the lift coefficient by approximately 0.11 according to thin aerofoil theory (Kuester, Intarapet and Borgoltz, 2018, p.1), highlighting that sub-degree positional accuracy as a direct aerodynamic data quality requirement. This review covers the literature behind the design of a motorised turntable for the UWE large subsonic wind tunnel, built as a quarter-scale proof-of-concept within the larger tunnel upgrade programme. Four areas are examined: existing wind tunnel positioning systems, motor selection, drive mechanism, and control system architecture. Each area links to a design decision – motor type, belt profile, gear ratio, and wireless protocol – where the literature either informed the choice or provides a benchmark to compare to.

3.2 Wind Tunnel Model Positioning Systems

Motorised turntables are an established solution across a range of scales and uses, with NASA's Oscillating Turntable (OTT) located in their Transonic Dynamics Tunnel at NASA Langley to university-built stepper motor solutions, like those used at Northumbria University and the University of Florida, both reviewed below.

NASA's OTT can pitch large semispan models up to 40 Hz, it is used to study flow conditions associated with flutter, limit cycle oscillations, and shock dynamics. The OTT demonstrates that precise, repeatable model motion is a fundamental requirement at the highest professional level, it was created specifically because there were no accurate flutter prediction codes available at the time (Piatak and Cleckner, 2003). The OTT is hydraulically actuated at a scale far beyond what is required of the new turntable at UWE, nor does it share the use case that would be required, it is not a benchmark the PoC is trying to compare to, but the core requirement of positional accuracy is shared.

Linehan, Shields and Mohseni (2014) document the design and validation of their Dynamic Actuation and Monitoring (DYNAMO) system; the system is capable of four degrees of freedom – pitch, roll, plunge and yaw – integrated into a tunnel with a test section of 0.61m x 0.61m with a maximum speed of 91.4 m/s. The yaw axis uses a turntable bearing between the base plate and rotating disc, actuated by a NEMA 42 stepper motor directly connected to the disc with a keyed hub, with a 500 CPR external encoder monitoring position. Performance of the yaw axis subsystem was limited by the load-to-inertia ratio of 140, which was acknowledged as being high for stepper motors; this meant that a large motor was required and limited dynamic tracking performance to frequencies below 1 Hz at an acceptable eRMS threshold of $< 10\%$. The inertia driven constraint on the direct drive configuration suggests that introducing a mechanical reduction between the motor and turntable would be advantageous, reducing the effective load inertia seen by the motor shaft by a factor of nine (gear ratio²), which allows for a smaller motor to be used. As shown in the DYNAMO control architecture (Linehan, Shields and Mohseni, 2014, Fig. 4) the motor's encoder is giving position feedback to the controller, rather than the motor driver – meaning the driver operates in an open-loop configuration. The proposed system improves upon this by using a closed-loop driver meaning that position errors are detected in real time and corrected. Like the proposed system the DYNAMO system incorporates LabVIEW for their control environment, confirming its suitability for this application.

A directly comparable system designed using stepper motor actuation, timing belt transmission and LabVIEW control is documented by Landeau et al. (2022). The system was designed to replace manual protractor-based AoA positioning with both static positioning control and oscillation modes, the same positioning problem addressed by the proposed system. Using a stepper motor with 1.8° of rotation per step and a transmission ratio of 1:1.8 they were able to achieve a theoretical resolution of 0.5° per motor half-step, however measured accuracy using an external encoder was 2% per

step, with an overshoot of 0.3° at high speed. Where Landeau et al. employ a T10 trapezoidal belt profile at a 1:1.8 ratio, the proposed system uses a GT2 curvilinear profile. The GT2 profile improves tooth engagement and reduces backlash compared to the T10 profile (AIMS Industrial, 2026), where the higher ratio provides greater torque multiplication and reduces effective load inertia at the motor shaft by a factor of nine compared to a factor of 3.24 for the Northumbria system. Unlike the system designed by Landeau et al. that uses an open-loop stepper, the proposed system uses a closed-loop stepper, this means that motor position will be verified by the internal encoder and any motor shaft positioning errors will be corrected in real time. Both systems measure accuracy under no aerodynamic load on the turntable, under realistic loads inaccuracies could be amplified so this must be taken into consideration. Again, like the DYNAMO system the Northumbria system opted to use LabVIEW as the control programme, this further solidifies LabVIEW as an established method for this application.

Ren (2020) documents the design of a pitch actuation system for the University of Manchester's Mach 5 supersonic tunnel. Although the system is designed for a different purpose to the proposed system parallels can be drawn from actuation and control decisions made. Ren evaluated hydraulic, pneumatic, wire-driven, and magnetic actuation before deciding upon electric hybrid stepper motors based on the accuracy and controllability - the same conclusion reached in the present work. Importantly, Ren identifies closed-loop control as essential under aerodynamic loading conditions, noting that an open-loop stepper was insufficient to guarantee positional accuracy against the aerodynamic forces experienced during testing. Like other systems assessed Ren (2020) also adopts LabVIEW as the chosen software to create a control environment, further establishing the software as an appropriate tool for the application required of the proposed system.

Across the systems reviewed stepper motor actuation along with a LabVIEW control environment emerges as the dominant academic approach. The proposed system shares the same architectural approach as Landeau et al. but improves upon the open-loop limitations identified across multiple systems, through closed-loop driver implementation. The GT2 belt reduction addresses the inertia problem identified by Linehan, Shields and Mohseni (2014).

3.3 Drive Mechanism Selection

As demonstrated by Linehan, Shields and Mohseni (2014) in the previous subsection, direct drive configurations present significant load-to-inertia challenges for turntable applications – a load-to-inertia ratio of 140 required a large NEMA 42 motor and limited dynamic tracking performance, highlighting that a mechanical reduction stage between the motor and turntable disc shaft is beneficial for motor sizing and positional accuracy. The drive mechanism directly determines achievable angular resolution, backlash, torque multiplication, and positional accuracy of the turntable system. A range of literature has been reviewed covering different use cases, assessing the benefits and downsides of each drive mechanism.

Kacalak et al. (2024) establish that worm gears are proven in precision engineering applications, such as machine tools and rotary tables – highlighting that it would be a suitable choice for the proposed system's drive mechanism. Their investigation identified backlash (the clearance between worm and wheel tooth surfaces) as the critical limitation of worm gear drives in precision positioning applications, finding clearance values ranging from 8 to $37\ \mu\text{m}$ in a precision worm gear, with clearance increasing as the tooth surfaces wear – meaning accuracy decreases over the systems operational life without proper maintenance. This correlates to the proposed systems application as it will be bidirectional and the backlash manifests as a dead zone at every direction reversal – the output shaft does not move until the tooth clearance is taken up, producing positional errors at every direction change. Kacalak et al. found that backlash can be reduced significantly, however the solution requires a patented mechanism with specialist manufacturing and assembly procedures, something that would not be available for a university level system. Although this paper highlights an inherent backlash issue with worm gear drives, the clearance values found are at such a small scale that the angular inaccuracies caused by them would be within a range that would be acceptable to ignore. On this basis, while worm gears are technically capable for this application, the backlash mitigation complexity and long-term maintenance requirements make them less suited for a university-level turntable system.

Budynas *et al.* (2015, pp.898–899) establish that timing belts transmit power at a constant angular velocity ratio, with no slip, require no initial tension and can achieve 97-99% efficiency, making them an attractive solution for precision drive requirements. AIMS Industrial (2026) provides a synchronous timing belt guide giving an overview of common belt profile families – trapezoidal, HTD curvilinear and GT2 modified curvilinear – noting that HTD tooth profiles have measurable backlash, impacting positional accuracy, while the GT2 profile offers tighter meshing reducing backlash. AIMS Industrial (2026) also lists the benefits of using synchronous belts over V-belts specifying that synchronous belts

have no slip, require no lubrication, up to 6% more efficient, and are less affected by dust and debris. Furthermore, timing belt drives are established in comparable wind tunnel applications, Landeau et al. employ a T10 timing belt in their system, and Aziz et al. (2008) who incorporated a belt drive for their automated AoA control in an educational wind tunnel.

Across the sources reviewed a timing belt driven mechanism appears the most appropriate transmission method for the accuracy requirements of the proposed turntable system – a complete comparison between transmission methods and final decision for the proposed system is conducted in Section 5.3.1.

3.4 Control System and Wireless Communication

As identified in Section 3.2, open-loop stepper systems employed in comparable wind tunnel positioning applications – specifically those documented by Linehan, Shields and Mohseni (2014) and Landeau et al. (2022) – rely on external position monitoring rather than real-time driver correction; Ren (2020) identifies this as insufficient under aerodynamic loading, where forces acting on the model risk causing step loss that an open-loop system cannot detect or correct. Kibenge (2018) identifies that open-loop stepper position error is proportional to loading – meaning motors are typically oversized by a factor of two or more to compensate for potential step loss, with no guarantee of reaching the target position under dynamic loading conditions. A closed-loop system addresses this by calculating motor current and phase angle in real time based on rotor position and compensates for position error resulting from frictional or inertial loads – this enables improvements in motor efficiency, reducing operating temperature and vibration, increasing the service life and reducing maintenance requirements on the motor. From the information gathered, implementing real-time position correction at the driver level rather than through an external control loop is identified as the appropriate solution for a turntable system operating under variable loading conditions.

Across the sources reviewed one thing remains constant, a LabVIEW control environment – Linehan, Shields and Mohseni (2014), Landeau et al. (2022), Ren (2020) and Aziz et al. (2008) all opted to use LabVIEW to control their systems, confirming that LabVIEW is the established platform for this application. However, it is worth noting that they all use LabVIEW in different ways, Linehan, Shields and Mohseni (2014) use NI PXI hardware with a LabVIEW interface, Landeau et al. (2022) use the ICNC library for driver communication, Ren (2020) uses LabVIEW alongside Arduino for motor command controls, and Aziz (2008) use LabVIEW for remote internet-based control – the range of use cases demonstrates that LabVIEW's flexibility accommodates different control architectures within the same platform. The present system similarly adopts LabVIEW as the control environment, with the control interface implemented as a SubVI to enable future integration into a wider data acquisition pipeline.

Wireless communication between the control PC and motor driver simplifies system installation and setup – once the turntable frame is installed beneath the test section and the motor and the receiver is powered, the only connection required from the control PC is a serial USB connection to the transmitter, eliminating the need to route a dedicated control cable between the test section and control room and satisfying the project requirement for control from outside the immediate tunnel area seen in Table 4.1. Eridani, Rochim and Cesara (2021) conducted an experimental comparison of three common short-range wireless protocols - ESP-NOW, Wi-Fi and Bluetooth – across range, transmission speed, latency, energy consumption, and barrier resistance using identical hardware. The key findings were that ESP-NOW consistently achieved the lowest and most stable latency of all three protocols tested – it operates as a peer-to-peer protocol, eliminating the need for IT department involvement in device registration or network configuration. A full protocol comparison and final decision can be seen in Section 5.4.1.

3.5 Summary

Across the literature reviewed, stepper motor actuation and LabVIEW control appears as the dominant approach for precision wind tunnel positioning at university-scale; independently validated by Linehan, Shields and Mohseni. (2014), Landeau et al (2022), Ren (2020) and Aziz et al (2008). Of those Landeau et al. (2022) and Aziz et al. (2008) employ a timing belt transmission, confirming it as the established drive mechanism for this application. The literature collectively identifies open-loop stepper actuation under aerodynamic load as the failure mode of comparable systems, as stated by Ren (2020) and evidenced by external monitoring architectures used by both Linehan, Shields and Mohseni (2014) and Landeau et al. (2022). Along with backlash in worm gear drives and load-to-inertia challenges in direct drive configurations as the key limitations informing the proposed design. Closed-loop stepper actuation with a timing belt reduction is identified across the reviewed literature as addressing these limitations. However, none of the systems reviewed verify model position under aerodynamic load, relying instead on motor encoder feedback only, when available

– Kuester et al. (2018) highlights that structural deflection and rotation of the model under full aerodynamic loading can cause angular offsets up to 0.11° , encoder-based measurement would not be able to detect this, as it is due the model itself and its mounting point deflecting, this represents a limitation shared by the present system that would require consideration at full-scale. The benchmarks and technical examples established in this review have directly informed the design specification and concept selection presented in Section 4 and 5.

4 Design Specification

4.1 Customer & System-Level Requirements

There was no formal customer brief set out for the design and manufacture of the turntable PoC - the requirements were derived from project objectives, dimensional constraints of the full-scale tunnel and a discussion with Dr. Samson Annapureddy, an aerodynamicist in charge of the tunnel upgrade at UWE. The requirements are as follows: the system must enable automated angular positioning of a wind tunnel model; it must operate within the physical constraints of the test section; it must be controllable from outside the tunnel; it must integrate with a LabVIEW control environment; it must be designed as a PoC with a clear pathway to full-scale implementation; and it must offer manual and automated control modes. The specific measurable targets derived from these requirements can be seen in section 0.

4.2 System Target Specification

#	Parameter	Requirement	Target Value	Justification
1	Angular range	360°	360°	Full experimental scope required for both yaw and AoA configurations
2	Angular resolution	2°	1°	Data quality requirement
3	Positional accuracy	Repeatable positioning	$\pm 0.3^\circ$	sub-degree accuracy required to minimise aerodynamic measurement error
4	Rotation speed	sufficiently slow to prevent model/turntable damage during repositioning	$\leq 10^\circ/\text{s}$ at turntable shaft	limit set via software to prevent unsteady aerodynamic loading on the model during repositioning
5	Control Interface	LabVIEW based, designed for future integration	Standalone LabVIEW VI with defined SubVI connector	LabVIEW is the established tunnel control environment; SubVI designed for future integration into wider tunnel control system
6	Control modes	Automated and manual control	Manual angle control and oscillation mode	Extends functionality beyond previous manual positioning methods
7	Scale	PoC	1/4 Scale	Budget constraints, PoC validates concept before purchase of components required for manufacture of full-scale design
8	Power supply	adequate power required for all components	Motor driver: 24V DC ESP32: 5V (USB)	CL42T driver operating spec ESP32 USB powered
9	Envelope	Fit within test section	Full-scale: disc $\varnothing$: 1005 mm, height 733 mm available. POC: disc $\varnothing$ 280 mm, height 190 mm	Dimensional constraints of UWE large subsonic wind tunnel high speed test section

10	Motor torque requirement	Motor must provide sufficient torque to accelerate the rotating disc assembly to the target angular velocity within the specified ramp time seen in Parameter 2.	$\geq 0.156 \text{ Ncm}$	From PoC inertia torque calculation (Appendix A)
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Table 4.1: System target specification matrix

4.3 Constraints & Standards

The only applicable standards identified for the PoC system were BS 7671 - the IET wiring regulations and is the primary standard covering all electrical installations that use mains power. The system also falls under PUWER 1998 as motorised work equipment in a university setting, specifically Regulation 11 (Dangerous parts of machinery), which requires guarding of the belt and pulley drive, and regulation 15 and 19 which require stop controls and power isolation capability – both addressed through the LabVIEW stop function and a mains switch respectively. For the full-scale implementation would require a more comprehensive PUWER compliance assessment, regarding emergency stop requirements (Reg. 16), structural stability anchoring (Reg. 20), and stricter guarding of rotating components. A full risk assessment conducted in accordance with UWE safety procedures is provided in Appendix E.

4.4 Risk Assessment

The risk assessment was conducted covering the assembly, installation, operation, and removal of the full-scale turntable system intended for use in the large subsonic wind tunnel rather than the PoC, which operates at low load with no aerodynamic loading. The main risks identified were injuries related to rotating components, electrical hazards and unexpected or uncontrolled actuation - the full risk assessment can be seen in appendix E.

5 Concept Development

5.1 System breakdown

The 1/4 scale proof-of-concept turntable system can be broken down into three different functional areas: the mechanical structure, the actuation system and the wireless control and LabVIEW interface.

The mechanical structure is made up from the frame, disc, bearings, shaft and shaft and bearing housings. The function of the mechanical structure is to support and locate the disc centrally, while transmitting rotational output from the actuation system.

The actuation system is made up from the stepper motor, driver, and belt drive assembly (including the shaft). The function of the actuation system is to generate and transfer the rotational motion to the turntable disc with the required resolution and accuracy.

The control system consists of the control PC, the two ESP32 microcontrollers, and the motor driver. The function of this system is to translate the operators commanded position from the control PC to data the motor driver can read and transfer into current to correctly pulse the motor.

5.2 Mechanical Structure

5.2.1 Frame Geometry

The frame geometry determines the stability, levelling capability, and manufacturability of the mechanical structure; three concepts were considered: triangular, square, circular, and octagonal.

A square profile frame would have had the benefit of having a larger enclosed area for components and it would have been an easy form to work with – requires no mitred cuts, while also matching the profile of the existing test section making attachment easier. The negatives of a square profile frame would be that it would require more material; a

different levelling system would have to be designed as four-point levelling would be over constrained; and it is less torsionally rigid than a triangular profile for the same material weight.

A circular frame would have the benefit of having no corners, meaning there would be no stress concentration at joints, and it would also have equal stiffness in all directions. The negatives of a circular frame are that it would be hardest to manufacture; it would have no flat faces naturally for component mounting; and it would be the most materially intensive of the three concepts.

An octagonal frame geometry has been identified as the preferred option for the full-scale system by Dr. Samson Annapureddy, due to the need to adequately support the 1005 mm disc around its perimeter – a triangular frame would not provide sufficient edge support at this scale. Additional benefits include a large enclosed area for component mounting and more fixing points to the floor. However, at the PoC scale it introduces further manufacturing complexity, and requires more material than other options.

The benefits of the triangular frame are that it is simple to manufacture, it is materially efficient, more torsionally rigid than a square frame and the three corners mean that it can be levelled easily. The negatives are that there is less enclosed area compared to a square frame of the same side length.

Criteria	Weight	Triangular	Square	Circular	Octagonal
Manufacturability	30%	7	8	4	6
Stability	30%	10	7	6	8
Levelling capability	20%	10	6	6	6
Material efficiency	20%	8	6	7	5
Weighted total		8.7	6.9	5.6	6.5

Table 5.1: Frame geometry concept down selection matrix

Note: scores are out of ten where a higher score indicates better performance against each criterion

Table 5.1 shows the down selection for the frame geometry; the triangular frame geometry scored highest overall, due to its superior stability and levelling capability, therefore it was selected. It should be noted that for the full-scale version an octagonal frame would be preferred.

5.2.2 Frame Material

The frame material determines the structural rigidity, manufacturability, and cost of the mechanical structure; two concepts were considered: timber and MDF construction and prefabricated aluminium extrusion profile.

The benefits of an aluminium extrusion frame construction are that they are much more dimensionally accurate and rigid, compared to a timber construction; easy to assemble with off the shelf connectors; and easier to scale to full-scale system. The downside of aluminium extrusion is that they are significantly more expensive than using timber.

A timber and MDF frame construction has the benefit of being low cost; easy to cut and work with basic tools; and it has sufficient strength for a PoC that will not be assessed under aerodynamic loads. The negatives of a timber frame are that it is less dimensionally accurate than machined extrusions and is not as rigid as an aluminium extrusion construction.

Criteria	Weight	Timber	Aluminium extrusion
Cost	35%	9	4
Manufacturability	30%	8	6
Ease of assembly	15%	5	9
Rigidity	15%	5	9
Scalability	5%	3	9
Weighted total		7.2	6.4

Table 5.2: Frame material down selection matrix

The down selection matrix for frame material can be seen in Table 5.2 – the highest scoring material was timber and MDF construction, for the lower cost and easier manufacturability, allowing iterations to be made easily during prototyping. It is worth noting that a timber construction will not be acceptable for the full-scale turntable system and aluminium extrusion construction would be the superior choice.

5.3 Actuation System

5.3.1 Transmission Method

The transmission method governs how the motor transmits its rotational movement to the turntable disc. Each method offers numerous benefits and downsides in cost, complexity, compactness, positional accuracy, and torque; the methods considered were direct drive, worm gear, and belt drive.

The benefits of the motor directly driving the disc are it is a simple arrangement – it does not require any intermediate components, there are no transmission losses, and it is compact. The downsides are there is no torque multiplication – the motor must provide the full torque required, and the resolution is limited to the motor step angle.

The benefits of a worm driven configuration are that high reduction ratios are achievable in a compact arrangement, worm gears are self-locking meaning it can hold position without power, and they can provide good torque multiplication. However worm gears are expensive compared to other transmission methods assessed; as discussed in section 3.3 they have inherent backlash issues, although angular accuracy would likely not be impacted enough to be an issue, they would require inspection and lubrication to reduce wear; they also offer lower efficiency than a belt driven configuration, and they are more complex to manufacture or source at a PoC scale.

A belt drive configuration using a GT2 belt profile is easy to source at a low cost, they have low backlash (Gates Corporation, 2014), and the pulley ratio provides torque multiplication and improves angular resolution by the same factor. The negatives of this configuration are that it requires more space than direct drive and that the belt will wear over time, requiring more maintenance compared to a directly driven system.

Criteria	Weight	Belt Drive (GT2)	Direct Drive	Worm Gear
Cost	25%	8	9	4
Positional accuracy	25%	9	5	7
Complexity & sourcing	20%	8	9	4
Torque multiplication	20%	8	1	9
Compactness	10%	6	10	8
Weighted total		8.1	6.5	6.2

Table 5.3: Transmission method down selection matrix

Table 5.3 shows how each method scored against the chosen criteria. A belt driven system scored the highest mainly for the low cost and lower complexity of implementation, while still offering improved positional accuracy and sufficient torque multiplication and was therefore selected as the transmission method for this system. Table 5.3: Transmission method down selection matrix

5.3.2 Actuation Method

The actuation method will determine the torque at which the disc is actuated, how accurately it is actuated, at what speed it is actuated, and how much the overall actuated system will cost. Four options were considered: DC motor with encoder, servo motor, open-loop stepper motor, and closed-loop stepper motor.

A DC motor and encoder actuated system benefits from high speed and smooth motion, while being low cost. However, it falls short in requiring a separate encoder and controller for position feedback; it has no holding torque – it requires power to hold position; a more complex control system is required, with PID tuning; and it is less precise at low speeds compared to the other options.

The benefits of a system actuated by a servo motor are that they are highly accurate, they have high torque and have a built-in encoder and closed-loop control – reducing complexity. The downsides of this approach are that servo motors and their drivers are inherently more expensive than the alternatives and they require power to hold position.

The pros of using an open-loop stepper motor configuration are that it would be simple to drive – there is no feedback loop required, they are low cost, they have good holding torque without power, and they are precise at low speed. The open-loop stepper loses out due to not being able to detect and correct lost steps, and they have resonance issues at low-mid speed – causing loss of torque, missed steps, uneven motion, resulting in vibration and noise (Zhou *et al.*, 2021).

A closed-loop stepper motor system has all the benefits of an open-loop stepper motor, while resolving their downsides. the CL42T driver used in this system also has a built-in anti-resonance compensation, further improving smoothness compared to an open-loop configuration (Stepper Online, 2023b), while also offering position feedback. The only downsides compared to an open-loop configuration are that they come at a slightly higher cost and complexity to configure.

Criteria	Weight	<u>Closed-loop stepper</u>	Open-loop stepper	Servo	DC motor & encoder
Positional accuracy	30%	9	5	9	6
Cost	25%	7	9	3	7
Holding torque	20%	9	9	3	3
Complexity	15%	6	6	6	4
Speed	10%	6	6	8	9
Weighted total		7.8	7.1	5.8	5.7

Table 5.4: Actuation method down selection matrix

The down selection matrix can be viewed in Table 5.4; the closed-loop stepper actuation method scored highest, this is primarily due to the high positional accuracy offered and holding torque advantages, whilst remaining relatively low cost. The open-loop stepper method came close but ultimately loses out, due to being unable to detect lost steps and correct them, so is unsuitable for the accuracy requirements.

5.4 Control System

5.4.1 Data Transfer Method

The data transfer method will determine how control commands are sent to the motor driver. The method must be capable of transmitting commands from outside the tunnel to the system mounted underneath the test section, wireless control would be preferred to reduce the need for cable routing and allow control to be done anywhere around the tunnel, not just at a specified location. Five methods were considered: manual hardware control, wired serial (USB), Bluetooth, Wi-Fi and ESP-NOW.

Manual hardware control benefits from its simplicity – no software is required, and there is no latency. However, it would not be possible to implement an oscillation mode without software; it would require the operator to physically be at the tunnel near where the control station is set up; positioning accuracy is dependent on operator skill – they must be able to properly read the control dials; and cable routing would be required.

The benefits of a wired serial connection are that it is reliable – no interference issues, low latency, and simple to implement. The downsides of this method are that cable routing would be required, operator is restricted to the length of wire and cable length can affect signal integrity.

Bluetooth data transfer benefits from low power consumption and that it would require no cable routing. While a range of 15 m (25 m with external antenna) would be sufficient for the current application given the control room proximity to the tunnel, Bluetooth loses out due to the highest latency compared to the other wireless protocols assessed (Eridani, Rochim and Cesara, 2021); and Bluetooth would require a more complex setup compared to ESP-NOW which operates natively on the ESP32 without additional configuration.

Wi-Fi data transfer benefits from the fastest transfer speeds of all the wireless protocols assessed, it has a large range (84 m) and has good barrier resistance. The downsides of a Wi-Fi connection are it has the highest power consumption compared to ESP-NOW and Bluetooth, it requires network infrastructure – a router and access point, and it can be susceptible to network congestion and interference (Eridani, Rochim and Cesara, 2021).

ESP-NOW has the greatest range at 185 m, increased to 220 m with an external antenna, the lowest latency of all wireless protocols assessed (Eridani, Rochim and Cesara, 2021), it requires no network infrastructure – it uses peer-to-peer connection, and it works on the ESP32 microcontroller used in the chosen concept. However, ESP-NOW falls short due to having a higher power consumption than Bluetooth (Eridani, Rochim and Cesara, 2021) and is only compatible with ESP32 and ESP8266 devices – meaning that if changes were made to the system and a different microcontroller was used, then the data transfer method would need to be reassessed.

Criteria	Weight	<u>ESP-NOW</u>	Manual	USB	Bluetooth	Wi-Fi
Cable routing	35%	9	2	2	9	9
Ease of setup	25%	8	7	8	5	4

Infrastructure required	15%	10	3	5	10	2
Latency	15%	9	10	9	4	7
Range	10%	9	2	3	4	7
Weighted total		8.9	4.6	5.1	6.9	6.2

Table 5.5: Data transfer method down selection matrix

Table 5.5 shows the down selection matrix for data transfer method. ESP-NOW scored highest due to requiring no additional infrastructure or cable routing and for being easy to set up with it being a native feature on the ESP32 microcontroller, and for those reasons was selected.

5.4.2 Control Interface

The control interface determines how the operator interacts with the system to input and monitor position commands; three options were considered: analogue controls, MATLAB and LabVIEW

An analogue control interface benefits from being simple, requiring no software and it would have no latency issues. However, it falls short due to be unable to have programmable modes without using software, and precision is based on dial resolution and operator skill.

The benefits of using a MATLAB interface are that it is a strong data processing and analysis application, it is good for algorithmic development, and it is widely used in academic research – meaning operators would already have experience using the software. The downsides are that it is less suited to real-time hardware control than LabVIEW – requiring additional toolboxes and more complex setup, and it is less intuitive for building operator interfaces.

LabVIEW benefits from being the industry standard for instrumentation and hardware control - it has native serial communication capability via VISA allowing direct interfacing with ESP32 transmitter, and NI DAQ integration allows for easy future incorporation into the wider tunnel control system, UI development is straightforward, it has real-time control capabilities built-in, and SubVI structure allows for modular design for future integration. The downsides of LabVIEW are it is limited when it comes to mathematical and algorithmic tasks compared to MATLAB, and it can be resource intensive on the control PC.

Criteria	Weight	<u>LabVIEW</u>	Analogue control	MATLAB
Programmable modes	35%	9	1	10
Hardware integration	25%	9	3	7
Ease of use	20%	8	8	7
Real-time control	20%	9	2	8
Weighted total		8.8	3.1	8.3

Table 5.6: Control interface down selection matrix

Table 5.6 shows how each control interface method scored against the chosen criteria. LabVIEW scored highest and MATLAB scored similarly, either option would be suited for the required function; however, LabVIEW was chosen due to its native DAQ integration and familiarity with the software for the required use case.

5.5 Selected Concept

The selected concept consists of a triangular timber and MDF frame to support the 280 mm MDF turntable disc. It is driven by a closed-loop NEMA 17 stepper motor whose rotation is transmitted to the central shaft through a 20-tooth pulley connected to a 60-tooth pulley on the central shaft – using a GT2 belt profile, giving a 3:1 reduction. Commands are sent from a LabVIEW control interface on an external PC, transmitted wirelessly using ESP-NOW between two ESP32 microcontrollers, the transmitter ESP32 is connected to the control PC through a serial USB connection; this sends commands to the receiver ESP32, which transfers the command to the CL42T driver, which generates the pulses required for the commanded movement. This concept was chosen as the best balance of performance and practicality within the PoC budget and manufacturing constraints, with material and component choices reflecting what was achievable at this scale rather than what would be used in the full-scale system.

6 Design Embodiment

6.1 Mechanical Structure Design

The mechanical structure consists of two horizontal equilateral triangular both with side lengths of 286 mm levels connected by six vertical pine members, with two members positioned at each corner of the triangle. The side length for the two triangle tiers was chosen as it needed to be large enough to fit the 200 mm lazy Susan bearing, which supported the turntable disc with a diameter of 280 mm. Two different types of timber make up the mechanical structure – 12 mm MDF and pine members with a profile of 32 mm x 32 mm.

The lower level is comprised of three mitred pine members with a length of 183 mm, joined together with PVA adhesive and supported with 30 mm wood screws, forming a triangle frame. A laser cut MDF triangle plate with a side length of 175 mm is glued into the frame as an infill to increase rigidity. A triangle base profile was chosen as it would allow for three levelling bolts to be placed in the corners to correct for any inconsistency in the lower construction.

Six pine members with a height of 85 mm are placed at the corners of the triangle frame to increase the height of the upper triangle base to allow the motor and pulley components to fit underneath. Each vertical member is secured from below using a 70 mm countersunk wood screw and PVA adhesive, with the screws flush to the underside of the lower frame. Three M8 T-nuts have been inserted into the top of each of the corner-most vertical members, creating a location for the levelling bolts to screw into. Note that only three vertical members are required to complete the structure, the three additional vertical members were added to accommodate 6 mm MDF side panels, which would have reduced twisting tendencies, this was later found to not be required.

The upper triangle level is made from laser cut MDF with a side length of 286 mm; each corner has an 8 mm drilled hole to fit the M8 levelling bolts. The bolts pass through the holes and engaged with the T-nuts below, secured with nuts and washers above and below, allowing adjustments to level the plate.

A 32 mm hole was drilled into the centre of the top triangle, with a 3D printed bearing housing inserted into the hole and secured with a friction fit. Two 608ZZ bearings were placed inside the housing also secured with a friction fit. The 200 mm lazy Susan bearing was secured into place using the pre-drilled mounting holes and $\varnothing$: 4 x 30 mm wood screws. The rotating disc rests on top of the lazy Susan bearing under its own weight and is centrally located by the central shaft passing through the shaft housing, ensuring the disc rotates level without drooping.

The disc is laser cut from MDF with a diameter of 280 mm and has a 16 mm hole drilled into the centre. A 3D shaft housing has been inserted into the hole, secured into place with a friction fit and $\varnothing$ 4 mm x 30 mm wood screws. A central shaft with a diameter of 8 mm is inserted into the bearings on the upper triangle plate and into the shaft housing, whilst being supported by an 8mm drilled hole at the base of the triangle infill, all secured with friction fits. On the section of shaft below the upper triangle plate, a 60-tooth pulley has been attached and secured with grub screws at a height of 72 mm above the lower MDF infill plate, level with the 20-tooth pulley attached to the motor shaft.

The motor is connected to the frame using a motor tensioner constructed from the same pine and MDF materials. The motor carriage is formed from three MDF pieces joined to make a U-shaped profile: the lower piece, where the motor sits, has a length of 140 mm and a width of 70 mm, and the two wall pieces each have a length of 140 mm and a width of 45 mm. Each wall piece is connected to the lower piece using two $\varnothing$ 4 mm x 30 mm wood screws and PVA adhesive. A fourth MDF piece is fixed across the top of the two walls with two 30 mm wood screws and PVA adhesive, closing the profile into a hollow rectangular section. This section is what connects the tensioner assembly to the frame, The U-channel remains open to accommodate the motor. The motor carriage allows the motor to be moved closer or further from the central shaft to adjust belt tension and is restrained by the tension hold plate.

Once the motor has been correctly tensioned, an additional MDF tension hold plate is secured to the carriage walls using two 30 mm wood screws to hold the motor at the correct position. To connect the tensioner assembly to the frame, an additional pine spacer piece with a length of 73 mm is fixed centrally on top of one of the mitred lower triangle members using two 40 mm wood screws and PVA adhesive. The closed rectangular section of the tensioner is then secured to the pine spacer using a single 70 mm wood screw and PVA adhesive securing the entire motor tensioner assembly to the frame. An additional two 60 mm long pine members were attached to the frame with 80 mm wood screws and PVA adhesive, placed on each side of the motor tensioner and secured with PVA adhesive, this was done to the tensioner from rotating.

6.2 Actuation System Design

The actuation system includes the motor, driver, and drive assembly. The stepper motor is a NEMA 17 stepper motor with an internal encoder, making it a closed-loop system. The motor has a holding torque of 0.42 Nm, a pull out torque of approximately 0.38 Nm (at operating speed of system) and a step angle of 1.8° (Stepper Online, n.d., 2023a). A NEMA 17 motor was chosen due to them being widely available at a low cost; the specific model was selected as the lowest cost option available, whilst being capable of meeting the torque requirements calculated in Appendix A.

The motor is driven by a CL42T closed-loop driver with an operating voltage of 24-48V, higher voltage would improve high-speed torque but 24V offers sufficient torque for the low operating speed of this system. The CL42T driver offers closed-loop correction, being able to detect lost steps from the encoder and correcting them automatically, without losing synchronisation. It also offers a soft start feature meaning when it is powered on there is no sudden jump, reducing mechanical shock. Additionally, the driver has fault outputs – it can output a signal when a fault is detected (Stepper Online, 2023b). The CL42T driver is configured with SW3 and SW4 set to the on position, giving a micro step resolution of 1600 pulses per revolution – giving a micro step of 0.225° at the motor shaft, 1600 pulses/rev was chosen against other micro step options available as it offers the smallest rounding error when commanding 1° increments, taking into account the 3:1 belt reduction. The S1 dial is set to position 4, giving a peak current of 1.4A (RMS 1A), this option also uses the factory default gain settings which the manual recommends (Stepper Online, 2023b). The CL42T driver was chosen as it was purchased in a set with the motor, ensuring encoder compatibility and was the most cost-effective closed-loop option available at the time of purchase. The CL42T driver meets all the functional requirements of the system including micro step resolution, anti-resonance features, fault output signalling, and soft-start feature.

The rotation from the motor shaft is transmitted to the central shaft – which rotates the disc – through a GT2 synchronous belt drive made up from a 20-tooth pulley on the motor shaft and a 60-tooth pulley on the central shaft, giving a 3:1 reduction ratio. The reduction ratio improves both the motor torque and the angular resolution, giving the shaft a holding torque and pull out torque of 1.26 Nm and 1.14 Nm respectively, far exceeding the calculated torque required in Appendix A. The angular resolution is improved to 0.075° per step at the turntable output.

An AS5600 magnetic rotary encoder was selected for validation testing of the system's accuracy. The encoder was designed to be mounted on the underside of the frame, with a 3D printed PLA magnet housing, holding a 6x3 mm disc magnet connected to the bottom of the shaft, which has been extended below the frame. The encoder has 4096 counts/rev, giving a resolution of 0.088° per count, which would enable validation of whether the shaft is being correctly rotated to the commanded angles and allow position error to be calculated. The AS5600 communicates via I²C through the SDA and SCL pins – the internal ZPOS register would be written to define the zero reference point, and the AGC and magnitude registers read to verify magnet positioning and field strength before validation testing. The encoder also offers the option to add a homing feature, enabling the disc to automatically find the 0° position, by reading the current angle on startup and returning to its set home position. A 3D printed shaft collar and PCB bracket were designed as the mounting solution, however printing could not be completed in time for submission – installation and validation testing of the AS5600 is therefore identified as a recommendation for further work in Section 8.4.

6.3 Control System

6.3.1 Hardware interface

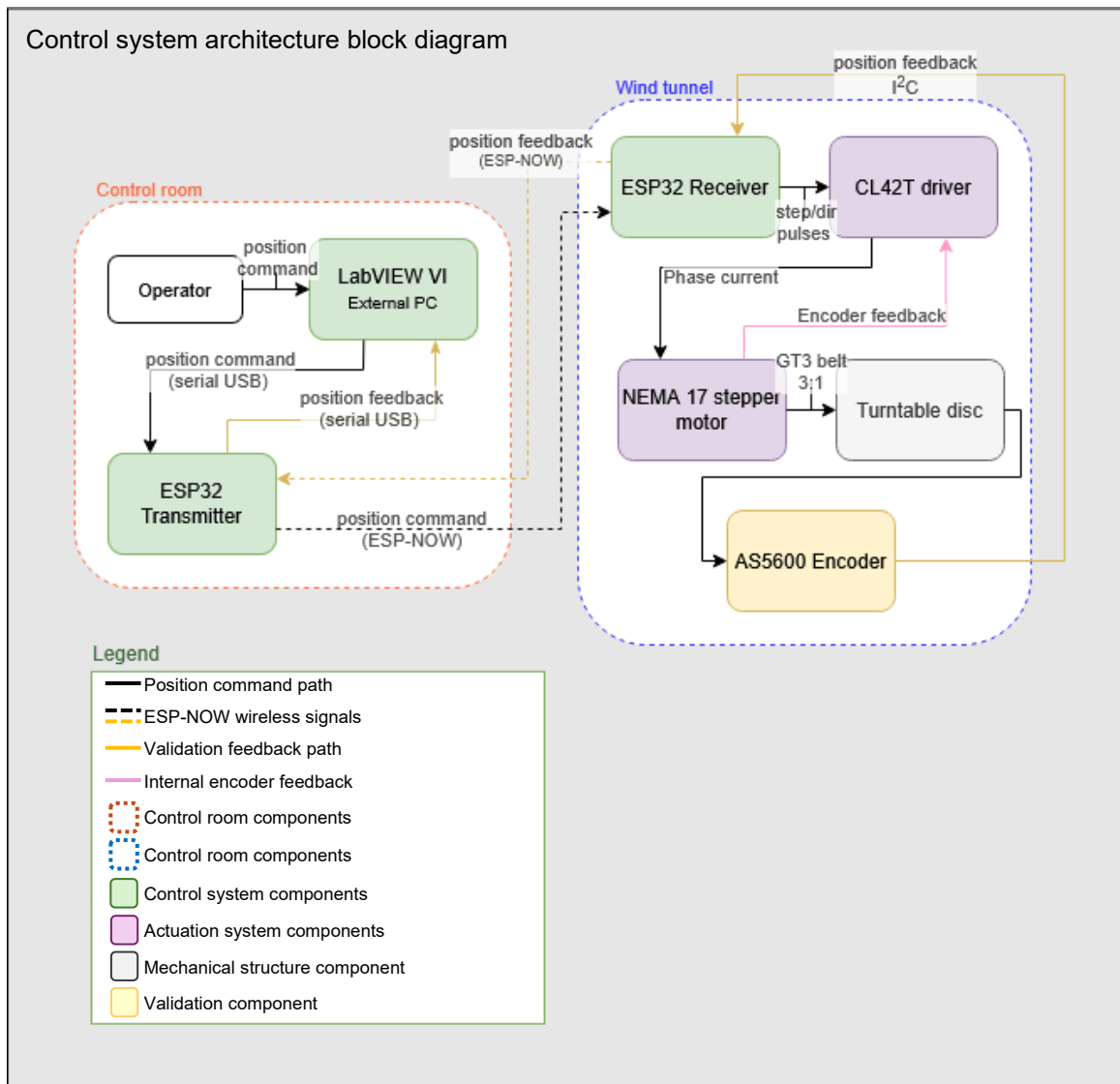


Figure 6.1: Turntable control system architecture block diagram

The control system architecture is shown in Figure 6.1 – the system operates across two environments: the control room - where the operator inputs position commands via LabVIEW on the control PC, and the wind tunnel - where the turntable motor and driver are located. The position command is sent to the ESP32 transmitter via a serial USB connection, the transmitter sends the position command through ESP-NOW – ESP's native wireless communication protocol, to the receiver ESP32. The receiver obtains the commanded position in the format angle,direction,speed followed by a newline character as a 9-byte struct variable, where angle is an integer value in degrees, direction is a binary value (0 = CCW, 1 = CW) with reserved values for homing (2) and empty queue (3), and speed is the desired motor speed in pulses per second. For the oscillation mode the receiver creates a queue of four commanded positions to prevent command overlap. The main loop continuously pulls from the queue and processes the next command only when the motor has completed its current move. The home and empty queue commands bypass or clear the queue immediately to ensure responsive control during stop and homing processes. The string is formatted in LabVIEW; no conversion is done by the transmitter; it is just sent to the receiver where it is then converted. The receiver sends the position command to the CL42T driver as a step and direction pulse format via two digital output pins: PUL+ - as a pulse signal, one pulse per micro step and DIR+ - as a direction signal, high = CW, low = CCW. The AccelStepper library in the receiver's code manages the conversion of these signals automatically, it calculates the required pulse timing based on the commanded speed and acceleration and toggles the PUL and DIR pins accordingly. To account for the 3:1 reduction between the motor shaft and disc shaft, the required pulses are multiplied by three, ensuring the disc is rotated to the correct angle. Once the required pulses have been received by the driver, the pulses are converted into two analogue current signals,

which then energise the two phases of the stepper motor. The motor shaft is then rotated, which transmits motion along the GT2 belt to the disc shaft, rotating the disc to the commanded position. Position feedback is provided by two encoders. The motor encoder feeds back to the motor driver, providing closed-loop correction internally, compensating for any position error and correcting it without intervention. The system was designed to include an AS5600 magnetic rotary encoder for validation. The AS5600 would have outputted the measured angle to the receiver ESP32 via I²C. The receiver would then send the value to the transmitter ESP32, through ESP-NOW, which would then transfer it to LabVIEW via serial USB, enabling validation of the system's accuracy in real time.

The hardware interfaces with the control PC to the motor driver through two ESP32 microcontrollers. The transmitter ESP32 is connected to the control PC through serial USB – the COM port assigned by windows must be verified in device manager before each session and updated in the LabVIEW VI if changed. Communication is done at a baud rate of 115200. The receiver ESP32 is connected to the CL42T driver via two digital output pins: pin 25 connects to the PUL-terminal and pin 26 connects to the DIR- terminal, with the ESP32's VIN pin connected to both PUL+ and DIR+ terminals. The CL42T driver is powered by a 24V DC power supply, whilst both ESP32 microcontrollers are powered by USB. The AS5600 encoder connect to the receiver ESP32 with SDA on pin 21 and SCL on pin 22, powered by 3.3V output from ESP32. The full wiring diagram can be seen in Appendix B.

6.3.2 Control VI Design

The LabVIEW VI is structured around a tab control with two operating mode tabs – manual and oscillate – with shared controls accessible regardless of the active tab screenshots of the front panel are provided in Appendix D.

The manual tab provides an angle input control accepting integer values from 0 to 360°, a direction toggle to select clockwise or anticlockwise rotation, and a send button to confirm the control input. Two LED indicators provide status feedback – the OK LED lights up when the command is successfully sent to the transmitter and the error LED illuminates when a fault is detected for example if the COM port is incorrectly defined in the block diagram or the serial connection is lost – both states can be seen in Figure D.3 and D.4 respectively.

The stop and home buttons are positioned outside the tab control and remain accessible from both operating modes. Pressing stop terminates the current operation and automatically returns the turntable to its 0° position. The home button sends a return to home command independent of the current operating mode but cannot be used if oscillation is in progress.

The block diagram is structured around an outer while loop containing the mode select case structure driven by the tab control value. VISA serial communication is configured outside the loop on startup, with VISA write nodes inside the loop transmitting the formatted command strings to the transmitter ESP32. The error cluster is chained through all VISA nodes ensuring any communication fault is communicated to the user and indicated on the error LED. The send button is connected to a case structure with the VISA write command chain located in the true case, and nothing in the false case, the same is true for the home command button. The manual angle control block diagram can be seen in Figure D.6.

The oscillate tab provides controls for centre angle, sweep amplitude, oscillation frequency in Hz, duration in seconds, and a start direction to define initial sweep direction. A calculated maximum frequency indicator updates dynamically with the chosen sweep amplitude, displaying the highest achievable oscillation frequency for the current motor max speed and acceleration. A warning LED illuminates if the chosen frequency exceeds the calculated maximum and the system will default to the calculated maximum – an example of what happens if chosen frequency is above the maximum possible can be seen in Figure D.5. A start button begins the oscillation sequence, if duration is set to 0 the sequence will run indefinitely until the stop oscillation button is pressed, which will end oscillation regardless of it has been commanded to run for a set time or indefinitely. A running status LED is lit green whilst the oscillation is in progress with an elapsed time display indicating how long the loop has been running.

The oscillate mode control block diagram is a bit more complex than the manual angle control – it is structured in four sequential stages, enforced by flat sequence structures to complete the correct execution order.

The first flat sequence contains two frames – the first sends the motor to the centre angle position, the second executes the first sweep, using the sweep amplitude value. Both frames use the calculated motor speed derived from the sweep amplitude and desired frequency.

Before oscillation begins the required motor speed and wait times are calculated from the operator inputs. The full arc pulse count is calculated as $2 \times \text{sweep amplitude} \times 4800 \div 360$, where 4800 is the pulses per disc revolution,

accounting for the 3:1 reduction. The required motor speed in pulses per second is calculated by dividing the pulse count by the half period in seconds. The wait time between commands uses a maximum node to select the largest of the desired half period and the actual motor move time calculated as $2 \times \sqrt{\text{pulses} \div \text{acceleration}(8000)} \times 1000$ – ensuring commands are never sent faster than the motor can execute them. Empirically testing found that 8000 pulses/sec² was the maximum acceleration that worked without overloading the motor with commands, causing inconsistency returning home. These calculations can be seen in figure D.7

The inner while loop then executes the remaining oscillation sweeps. Inside the loop is a case structure, which alternates the direction using a Boolean shift register – the shift registers flips on every iteration via a not gate gate, ensuring each sweep reverses direction. The format into string node builds the command string using 2 x sweep amplitude as the angle, the alternating direction value, and the calculated speed. An elapsed time node monitors the session duration, when the set duration expires, or the stop button is pressed the loop exits.

After the while loop exits a second flat sequence handles shutdown – the first frame sends a clear queue command to clear the receiver queue and waits for the current motor move to complete, the second frame sends the home command returning the turntable to 0°.

Finally, outside all case structures, the VISA write node for the home position and VISA close node execute, closing the serial port cleaning and returning motor home in the case that the operator hasn't done so themselves. The full block diagram can be seen in Figure D.8

6.3.3 Control Firmware

The ESP32 firmware is written in C++ using the Arduino framework. The transmitter firmware takes in serial commands received by LabVIEW in the format angle,direction,speed\n, and forwards them to the receiver via ESP-NOW. The receiver firmware implements a buffer queue of eight command slots, ensuring commands are not lost during rapid sequential transmission during oscillation control mode. Commands are taken out the queue in the order they were received and only processed when the motor has completed its current move, stopping command overlap. The AccelStepper library manages both pulse generation and acceleration profiling for the motor. Full code for both transmitter and receiver is provided in Appendix D.

6.3.4 Design for Future Integration

The LabVIEW VI is structured as a SubVI with a defined connector pane, allowing it to be embedded into a wider LabVIEW DAQ without modification to the control architecture. This directly supports the planned integration with the digital twin sub-system currently in development, where sensor data from all sub-systems will be collected through LabVIEW and transferred to MATLAB for real-time computation.

The AS5600 magnetic rotary encoder once installed will provide position feedback through the receiver ESP32 to transmitter, then control PC, allowing operator to verify the actual disk angle. Using the encoder, it would also be possible to create an automatic homing system, meaning that the system would no longer need to be manually homed by hand if it loses position.

6.3.5 Safety Features

At the hardware level, the CL42T driver's enable pin can be used to prevent motor motion independently from the control system without power down the driver, and the ALM pin outputs a fault signal when a position following error or overcurrent condition is detected.

At the software level, the stop button is accessible from all operating modes and stops motion immediately. The error LED provides visual feedback if serial connection is lost or COM port is not correctly configured.

7 Operational Considerations

7.1 Operating Sequence

1. Power on the CL42T driver and ESP32 microcontrollers – the CL42T driver is powered by a 24V DC power supply and the ESP32 microcontrollers are powered via USB. No specific power-on order is required.

2. Open LabVIEW on control PC – serial USB connection must be checked to ensure that LabVIEW is reading the correct COM port for transmitter, the receiver doesn't need to be put in a specific port as no data is being sent to it through serial USB, it just needs to be powered on.
3. **[only applicable once the AS5600 has been installed]** VI checks AS5600 AGC and magnitude registers – this confirms encoder is operational and magnet is correctly positioned, if AGC or magnitude values are out of range, the encoder mount should be checked before proceeding.
4. System is ready for operation.
5. Operator chooses a control mode in the tab select on LabVIEW – either manual or oscillation.
6. Operator inputs desired parameters and executes the control.
7. System moves to commanded position.
8. To shut down operator must press stop button – the system automatically returns to 0° home position.
9. Power down driver and microcontrollers

It is important to note that if the microcontrollers or driver is powered down before stop button is pressed the system will not be able to return home automatically and it must be manually positioned to 0° home position by hand with the driver powered down, before its next use.

7.2 Validation Testing

Due to the AS5600 encoder mount not being completed, formal accuracy against the $\pm 0.3^\circ$ specification was not possible. Instead function verification was carried out to verify the system worked correctly across all its modes.

Manual angle control was validated by commanding a single angle repeatedly – for example, 1° commanded 360 times – and observing whether the disc returned to its original position, verified visually against a reference marking on the shaft, with a 0.5° buffer either side. This was repeated across a range of angles. Any position error from individual commanded movements would accumulate over repeated cycles and they would be visually noticeable; it is reasonable to conclude that the system has an accuracy of approximately $\pm 0.5^\circ$, as no accumulation was visually noticeable across all tests.

Homing functionality was validated by commanding a sequence of 50 random angle moves across various directions and verifying the automatic home function returns to 0° visually, based on a reference marking on shaft. The system successfully returned the home reference in all test runs.

Oscillation mode was validated by running sweeps at various centre angles, sweep amplitudes and frequencies, and observing consistent and equal bidirectional rotation. At lower frequencies, a stopwatch was used to time complete oscillation cycles and verifying the frequency against the set value – at lower frequency measured frequency were consistent with commanded values. Timing verification was not possible at higher frequencies due to the speed of motion.

Each test was repeated three times, testing tables can be found in Appendix F.

7.3 Performance Against Specification

Specification	Target	Achieved	Status
Angular resolution	1°	1° (lower is possible, limited in software by design)	Met
Positional accuracy	$\pm 0.3^\circ$	$\pm 0.5^\circ$ visually validated (formal validation pending AS5600 installation)	Partially met
Operating range	360°	360°	Met
Max operating speed	$\leq 10^\circ/\text{s}$	$20^\circ/\text{s}$ demonstrated safely during testing	Exceeded

Oscillation mode	Required	Implemented	Met
Wireless control	Required	Implemented	Met
Automatic homing	Required	Implemented	Met

Figure 7.1: Performance against specification criteria table

7.4 Reliability & Failure Modes

The most significant failure identified was the shaft housing connection between the disc and drive shaft, which was found to be incapable of holding the disc, due to manufacturing tolerances in the 3D printed housing resulting in an oversized shaft connection hole and the bore screws being incapable of gripping shaft. This could be resolved by filing a flat section on the shaft where the interfaces with the grub screw and reprinted with a tighter shaft hole, or by applying adhesive to the shaft, which would result in a permanent connection, which would not be preferred.

A mechanical failure mode was identified with the lazy Susan bearing connection to the turntable disc – the bearing was not rigidly fixed to the disc, resulted in sufficient contact between the bearing surface and the disc underside. Along with the higher-than-expected friction of the bearing rotation caused the disc to rotate freely above the bearing rather than being supported by it. This can be resolved in future iterations by using a lower friction bearing and by drilling holes in the disc where it contacts the bearings connections points ensuring the disc slots in.

A command queue overflow failure mode was identified during oscillation mode testing at higher frequencies, commands were sent faster than the motor could execute them, causing inconsistent motor velocity and unreliable homing. This was prevented by implementing a maximum frequency calculation in LabVIEW.

A homing position drift failure mode was identified over long oscillation sequences, minor timing variation between commanded and executed moves caused small errors to accumulate, this would cause unreliable homing. This was fixed in the receiver ESP32 firmware by not allowing the next movement to occur until the motor has completed its current movement. This may have caused inaccuracies in frequency timing at higher frequencies but cannot be tested until the AS5600 encoder has been implemented.

COM port reassignment by Windows when the ESP32 is reconnected each session requires the operator to verify and update the COM port in LabVIEW VI block diagram before each session, as detailed in operating sequence in Section 7.1.

7.5 Maintenance & Calibration

The PoC turntable system requires minimal maintenance during its operational lifecycle. The GT2 belt tension should be checked periodically and adjusted using the tensioner if belt shows signs of slacking. The belt and pulley teeth should be inspected periodically for wear and debris also.

The 608ZZ bearings and lazy Susan bearing require no lubrication as they are sealed units. If bearing roughness is detected, they should be replaced.

If home position is out of place, it should be recalibrated, in its current state the system requires manual hand positioning of the disc. If the disc is not at its 0° reference point when the system starts the home position tracking will be off as it calculates home position based off where it began when it was powered up. With the installation of the AS5600 the manual calibration step will be replaced with an automated homing configuration using the encoder zero reference.

The COM port assignment should be verified in device manager at the start of each session.

7.6 Future Scalability to Full-Scale

This section identified key considerations for scaling the present PoC turntable system to a full-scale installation within the UWE large subsonic wind tunnel.

At full scale, the turntable disc will interface with a six-component force balance mounted between the plate and the model being tested. The present PoC disc does not include dedicated balance mounting points – however the standard MDF construction, which is likely to be carried over to the full-scale version, allows for straightforward retrofitting of mounting hardware in future iterations.

Additionally, full scale integration into the wind tunnel will require consideration of the tunnel floor mounting, structural rigidity under loading, and compatibility with test section and balance dimensions. For the full-scale version, it is recommended that the frame be constructed from prefabricated aluminium extrusion profiles, they offer a high strength-to-weight ratio, modularity and T-slot connectivity simplifies frame assembly and future modification. Recommendation for how the frame should be mounted to the floor depends on whether the turntable is needed as a permanent or removable installation. For a permanent installation, chemical anchors are recommended – this method uses threaded rods bonded into drilled holes in the concrete floor using two-part epoxy, providing higher load capacity than alternatives methods by distributing load across the full bonded depth of the hole rather than at a single expansion point. Furthermore, chemical anchors are resistant to vibration-induced loosening, which is likely as it will be installed near the tunnel fan. If removal of the turntable is required it is recommended that expansion anchors should be used instead, they provide sufficient rigidity and load capacity while allowing clean removal by unbolting without damage to the floor.

The motor and drive in the PoC versions are sized for PoC loads, at full scale with a larger disc, higher aerodynamic loads, and the force balance and model mounted the motor and drive would need to be resized; the calculations for required central shaft torque can be seen in Appendix A, it is therefore recommended that a motor capable of outputting 6.18 Nm of pull out torque be used in combination with a 5:1 reduction this will provide adequate torque to rotate the components under load, based from the calculation in Appendix A.

The proposed turntable system will eventually interface directly with the digital twin sub-system currently being developed by (Hendry, 2026). The current digital twin comprises of a visualisation designed on Unity and mathematical model implemented on MATLAB; in its future state, live sensor data including model orientation from the wind tunnel turntable will be acquired through LabVIEW and transferred to MATLAB for real-time aerodynamic coefficient computation and mathematical model validation. This integration requirement informed the decision to implement the turntable control interface as a LabVIEW SubVI, allowing for easy integration into the wider data acquisition pipeline without modification to the control architecture.

8 Conclusions

8.1 Summary of Design

This report covered the design and implementation of the quarter-scale proof-of-concept turntable system for the large UWE subsonic wind tunnel. The system consists of a timber frame supporting a 280mm MDF disc, driven by a closed-loop NEMA 17 stepper motor through a GT2 3:1 timing belt reduction. Position commands are sent from a LabVIEW control interface on an external PC, transmitted wirelessly through ESP-NOW between two ESP32 microcontrollers, and executed by a CL42T closed-loop stepper motor driver.

Two operating modes were implemented: manual and oscillation control, along with a homing feature. The control system architecture was designed with future integration in mind, with the LabVIEW VI structured as a SubVI and a position feedback path implemented for future AS5600 implementation.

8.2 Specification Compliance

The system meets the majority of the target specification defined in Section 4.2. The 1° resolution target is met. The full 360° range is achieved. The maximum operating speed exceeded the original 10°/s set out in the original specification, with reliable operation demonstrated at 20°/s, however this may need to be reassessed for the full-scale version as it may cause inertia related issues.

The positional accuracy specification criterion of $\pm 0.3^\circ$ was not validated due to the AS5600 encoder mount not being completed within the project timeline. Based on visual validation the system achieves approximately $\pm 0.5^\circ$, which falls short of the target; however, this figure is limited by the resolution of visual observation and encoder-based validation may demonstrate improved accuracy.

The wireless control requirement is met through the ESP-NOW implementation. The manual and oscillation mode and homing requirement is also met.

Overall, the control system and drive mechanism perform as designed, with the outstanding specification being positional accuracy validation, which requires the completion of the encoder mount, as detailed in Section 8.4.

8.3 Areas of Concern / Limitations

The primary area of concern is the inability to properly validate positional accuracy against the $\pm 0.3^\circ$ criterion. Without the AS5600 encoder validation, the true accuracy of the system remains unconfirmed, and the visual validation method used is insufficient to draw a definitive conclusion about whether the criterion has been met.

The mechanical issues identified during testing, specifically the shaft housing and lazy susan bearing connections, highlight the limitations of timber and 3D printed construction for precision engineering applications. Manufacturing tolerances inherent to these methods introduce variability in dimensions that would not be present in machined metal assembly and represent the primary risk to the system meeting its accuracy specification, as any warp or an off-centre hole in the MDF disc would transfer inaccuracy into the system.

The system has not been tested under any load – all validation was carried out with the shaft unloaded. Performance under realistic wind tunnel flow regimes remain unverified.

8.4 Recommendations for Further Work

The following recommendations are made for the future development of the turntable system:

The most immediate recommendation is the installation and complete configuration of the AS5600 encoder. The 3D printed shaft collar for the magnet has been printed and installed, and the design of the mount has been completed and requires printing and testing. Once installed the encoder will allow validation of the positional accuracy against the accuracy criterion and provide position feedback and automatic homing capabilities.

The shaft housing connection between the disc and drive shaft should be resolved by filing a flat section onto the shaft where it contacts the grub screw. The disc should also have mounting holes drilled so it can properly interface with the lazy Susan bearing, to prevent free rotation.

The system should be tested under scaled aerodynamic load to verify performance against realistic operating conditions. The testing should include positional accuracy and oscillation consistency.

For the full-scale system it is recommended that the frame be constructed from aluminium extrusion profiles and that the motor and drive be resized based on the full-scale torque calculations provided in Appendix A.

Finally, a random angle mode is recommended as a future addition to the LabVIEW control interface, allowing the turntable to be commanded to random angles within a defined range – this increases the use cases of the turntable system, by replicating the random nature of wind direction changes.

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11 Appendices

11.1 Appendix A — Motor Sizing Calculation

All known values and calculations are done to three significant figures.

a) PoC Torque Requirement Calculation

Assumptions:

- Disc is modelled as a solid uniform cylinder – central hole ignored
- Disc mass taken from direct measurement
- Maximum angular velocity and target acceleration time taken from system target specification (p.5)
- Bearing and belt friction not calculated, instead absorbed by safety factor
- Aerodynamic loading not considered

Known values:

Disc mass: $m = 0.454 \text{ kg}$

Disc radius: $r = 0.14 \text{ m}$

Max angular velocity: $\omega = 10^\circ/\text{s} = 0.175 \text{ rad/s}$

Target acceleration time: $t = 1 \text{ s}$

Belt reduction ratio: $n = 3:1$

Safety factor: $SF = 2.0$

1. Moment of inertia of disc (I)

$$I = 0.5 \times m \times r^2$$

$$I = 0.5 \times 0.454 \times 0.14^2$$

$$I = 0.00445 \text{ kg} \cdot \text{m}^2$$

2. Required angular acceleration (α)

$$\alpha = \omega \div t$$

$$\alpha = 0.175 \div 1$$

$$\alpha = 0.175 \text{ rad} \cdot \text{s}^{-2}$$

3. Required torque at turntable (T)

$$T = I \times \alpha$$

$$T = 0.00445 \times 0.175$$

$$T = 7.79 \times 10^{-4} \text{ Nm}$$

4. Including safety factor (accounts for friction, manufacturing variability and general uncertainty)

$$T_{\text{required}} = T \times SF$$

$$T_{\text{required}} = 7.79 \times 10^{-4} \times 2$$

$$T_{\text{required}} = 1.56 \times 10^{-3} \text{ Nm} = 0.156 \text{ Ncm}$$

B) Full-Scale Torque requirement calculation

Assumptions:

- Full scale disc modelled as a solid cylinder
- Disc mass calculated from measured MDF density of PoC disc (622 kg/m³)
- Balance modelled as solid cylinder using manufacturer specified mass for 6A154C balance
- Model mass taken at max of 4 kg, modelled as a point mass at disc edge
- Max angular velocity and acceleration time taken from system target
- Bearing and belt friction not calculated, absorbed by SF
- Aerodynamic yaw load taken as max rated yaw moment of 6A154C balance

Known values

- Disc outer radius: $R = 0.5025 \text{ m}$
- Disk thickness: $h = 0.012 \text{ m}$
- MDF density: $\rho = 622 \text{ kg/m}^3$
- Model mass: $m_{\text{model}} = 4 \text{ kg}$
- Balance mass: $m_b = 2.1 \text{ kg}$
- Balance radius: $r_b = 0.077 \text{ m}$
- Max angular velocity: $\omega = 20^\circ/\text{s} = 0.349 \text{ rad/s}$
- Target accel time: $t = 1 \text{ s}$
- Safety factor: $SF = 1.5$
- Max yaw moment: $M_z = 20 \text{ Nm}$

1. Disc mass (m_{disc})

$$\begin{aligned}V &= \pi \times R^2 \times h \\V &= \pi \times 0.5025^2 \times 0.012 \\V &= 0.00952 \text{ m}^3 \\m_{\text{disc}} &= \rho \times V \\m_{\text{disc}} &= 622 \times 0.00952 \\m_{\text{disc}} &= 5.92 \text{ kg}\end{aligned}$$

2. Moment of inertia of disc (I_{disc})

$$\begin{aligned}I_{disc} &= 0.5 \times m_{disc} \times R^2 \\I_{disc} &= 0.5 \times 5.92 \times 0.5025^2 \\I_{disc} &= 0.747 \text{ kg} \cdot \text{m}^2\end{aligned}$$

3. Moment of inertia of model (I_b)

$$\begin{aligned}I_b &= 0.5 \times m_b \times r_b^2 \\I_b &= 0.5 \times 2.1 \times 0.077^2 \\I_b &= 0.00623 \text{ kg} \cdot \text{m}^2\end{aligned}$$

4. Moment of inertia of model (I_{model})

$$\begin{aligned}I_{model} &= m_{model} \times R^2 \\I_{model} &= 4 \times 0.5025^2 \\I_{model} &= 1.010 \text{ kg} \cdot \text{m}^3\end{aligned}$$

5. Total moment of inertia (I_{total})

$$\begin{aligned}I_{total} &= I_{disc} + I_b + I_{model} \\I_{total} &= 1.76 \text{ kg} \cdot \text{m}^2\end{aligned}$$

6. Required angular acceleration (α)

$$\begin{aligned}\alpha &= \omega \div t \\ \alpha &= 0.349 \div 1 \\ \alpha &= 0.349 \text{ rad} \cdot \text{s}^2\end{aligned}$$

7. Required inertial torque at disc shaft ($T_{inertia}$)

$$\begin{aligned}T_{inertia} &= I_{total} \times \alpha \\T_{inertia} &= 1.763 \times 0.349 \\T_{inertia} &= 0.615 \text{ Nm}\end{aligned}$$

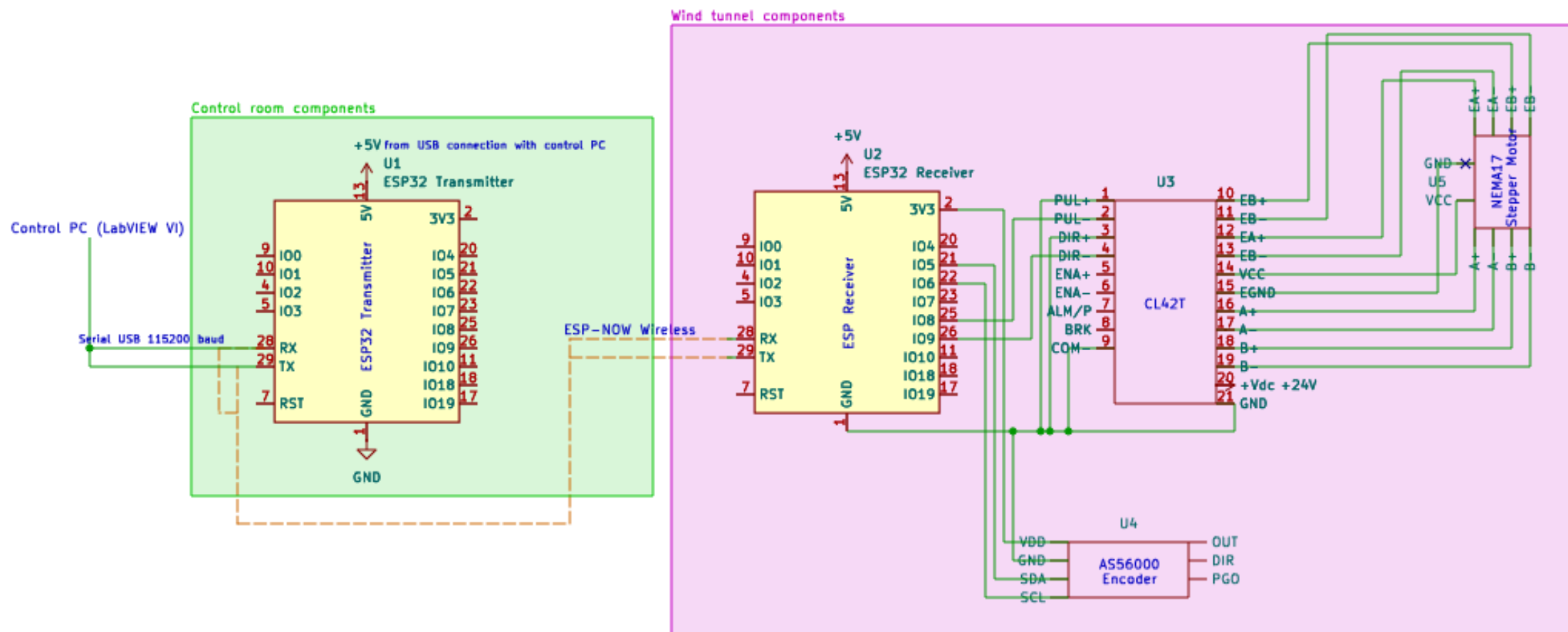
8. Total torque at disc shaft including aerodynamic load (T_{total})

$$\begin{aligned}T_{total} &= T_{inertia} + Mz \\T_{total} &= 0.615 + 20 \\T_{total} &= 20.6\end{aligned}$$

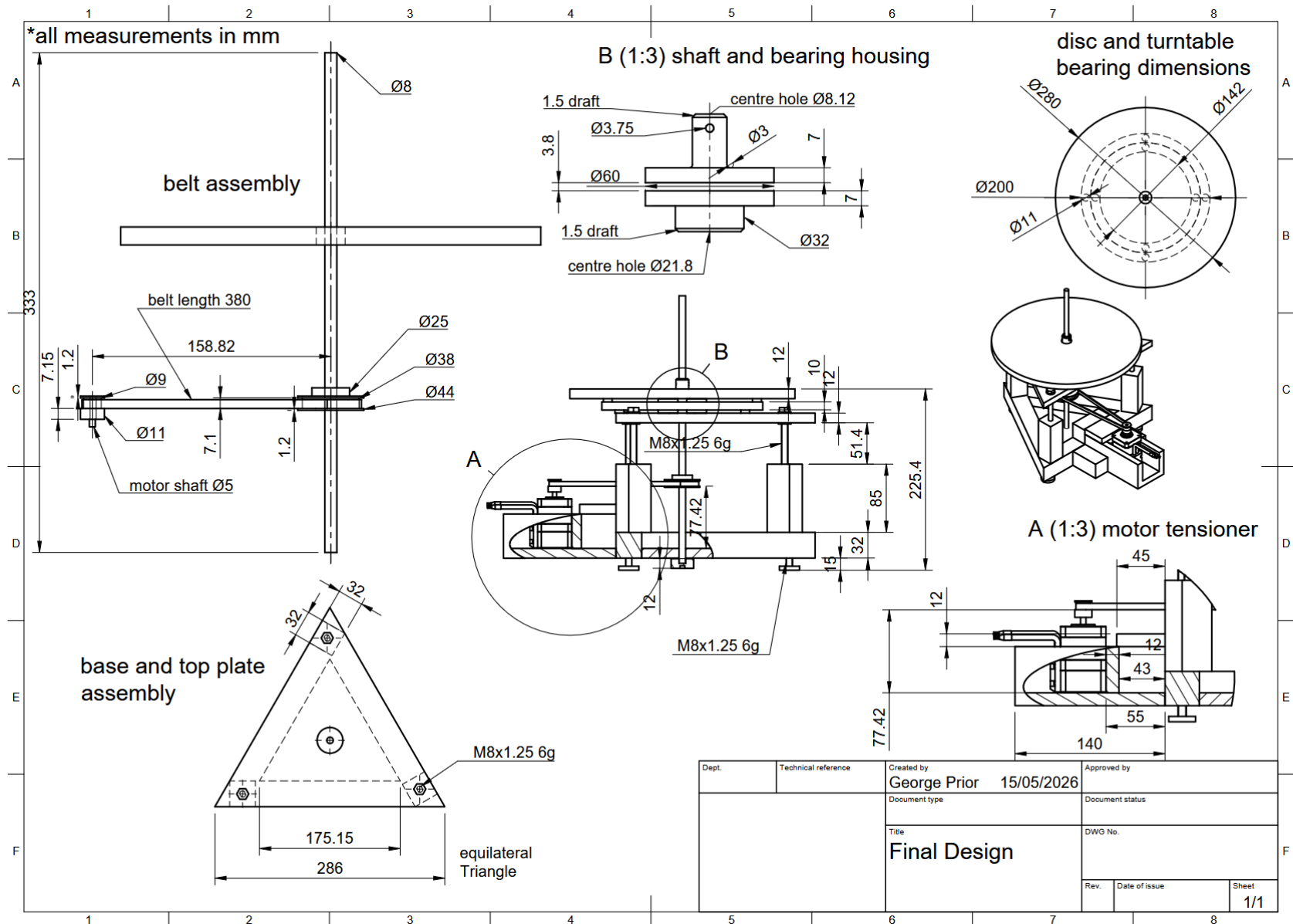
9. Including safety factor ($T_{required}$)

$$\begin{aligned}T_{required} &= T_{total} \times SF \\T_{required} &= \mathbf{30.9 \text{ Nm}}\end{aligned}$$

Appendix B — Electrical Wiring Diagram



11.2 Appendix C — Engineering Drawings



11.3 Appendix D — LabVIEW VI Screenshots and ESP32 C++ Firmware Code

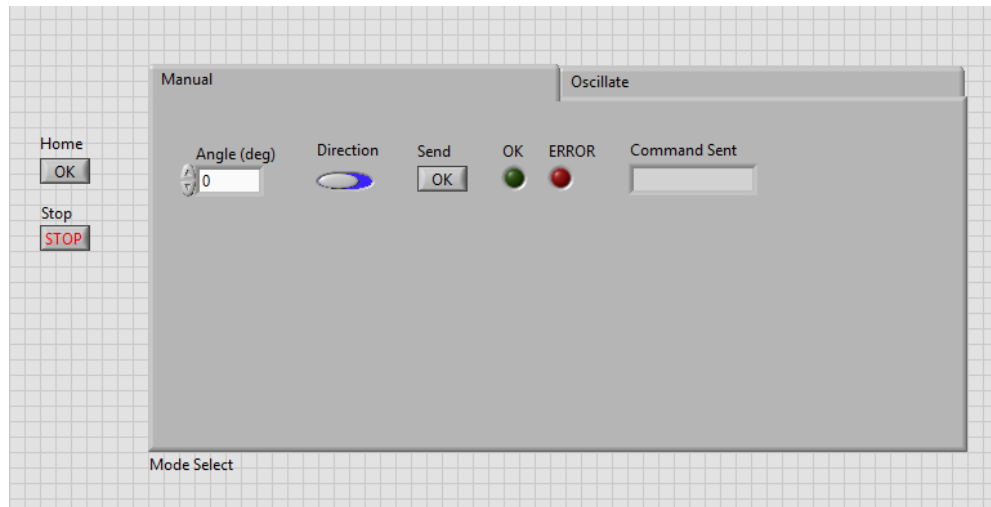


Figure D. 1: LabVIEW manual angle control interface

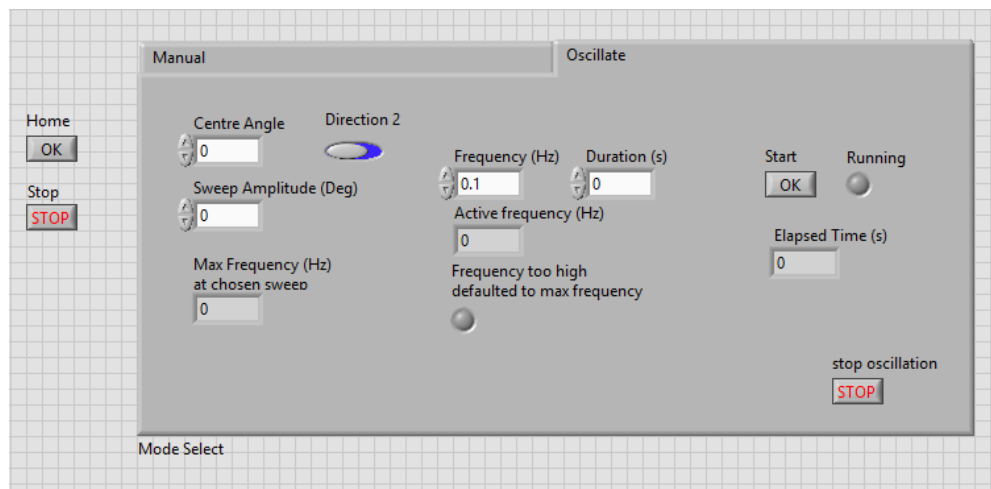


Figure D. 2: LabVIEW oscillation control interface

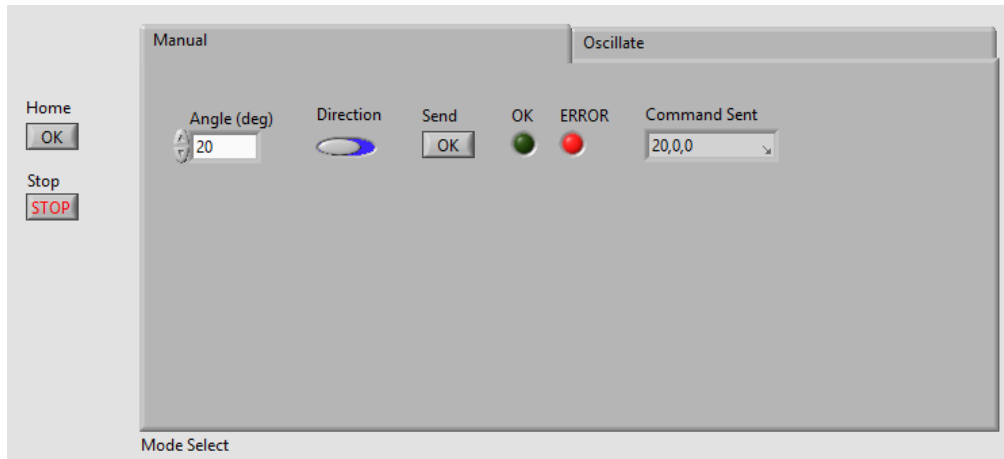


Figure D. 3: Manual angle control error example

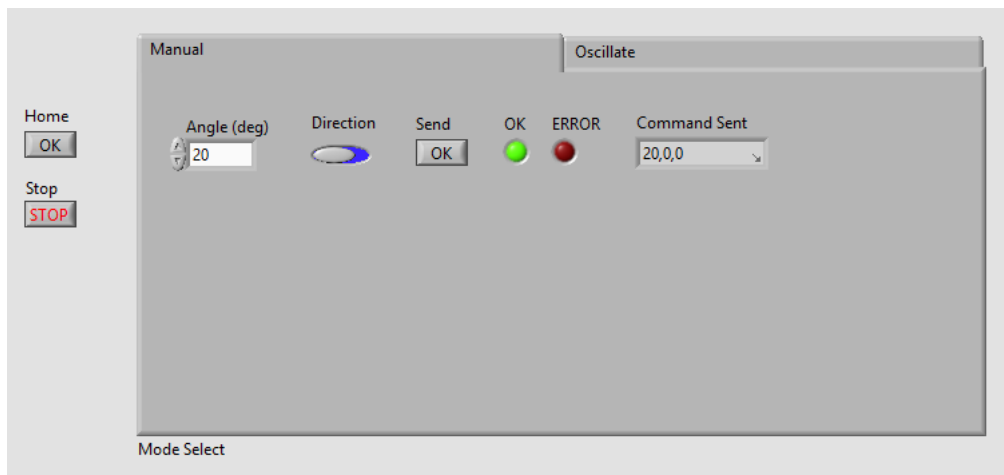


Figure D. 4: Manual angle control no error

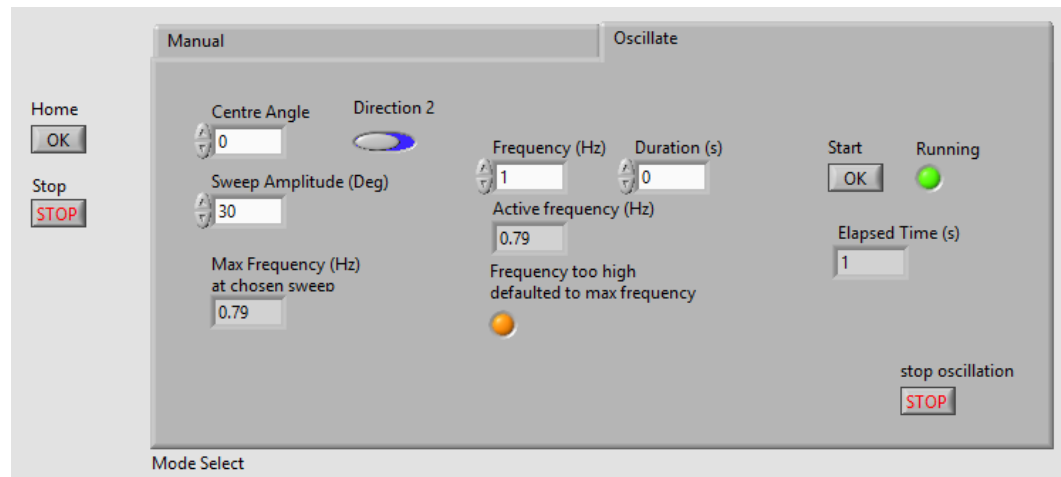


Figure D. 5: Oscillation control max frequency control

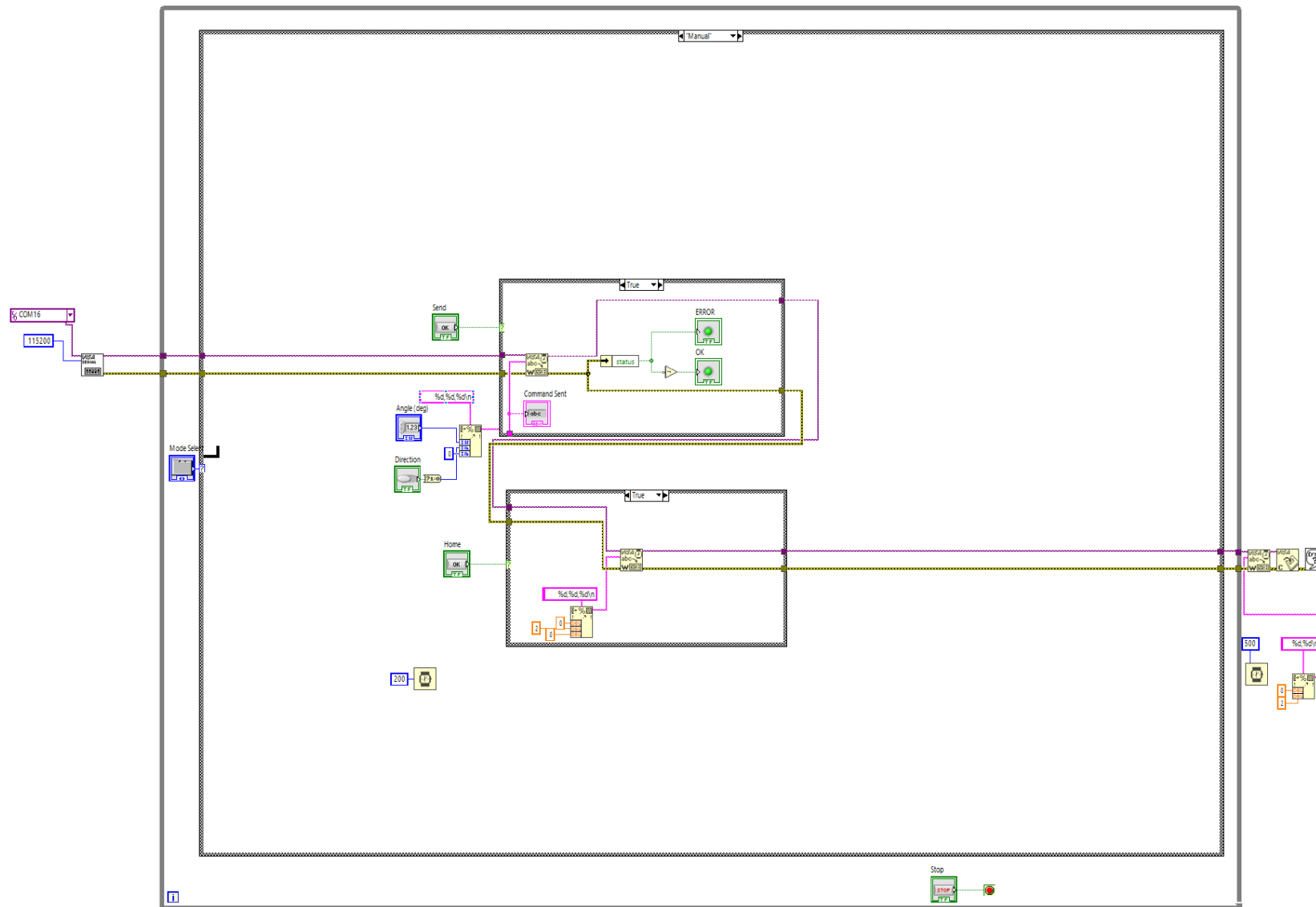


Figure D. 6: Manual control block diagram

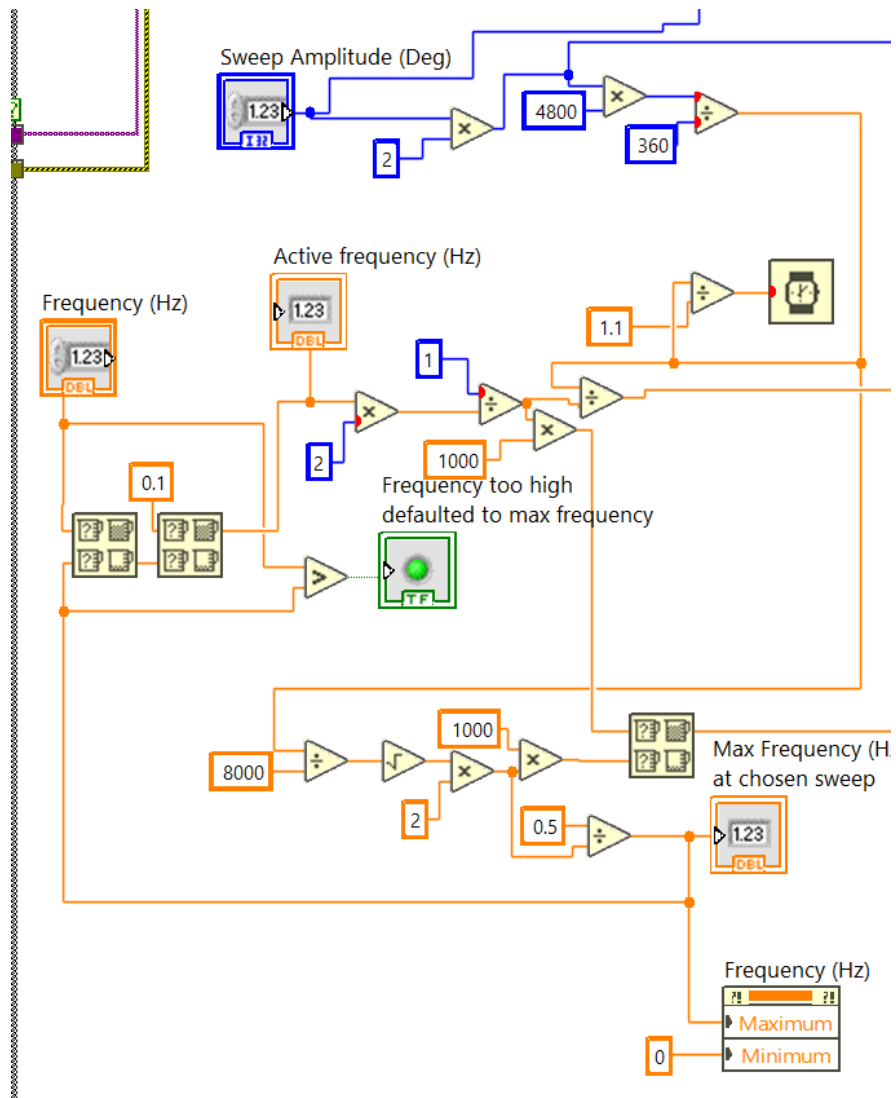


Figure D. 7 Oscillation control frequency and speed calculation

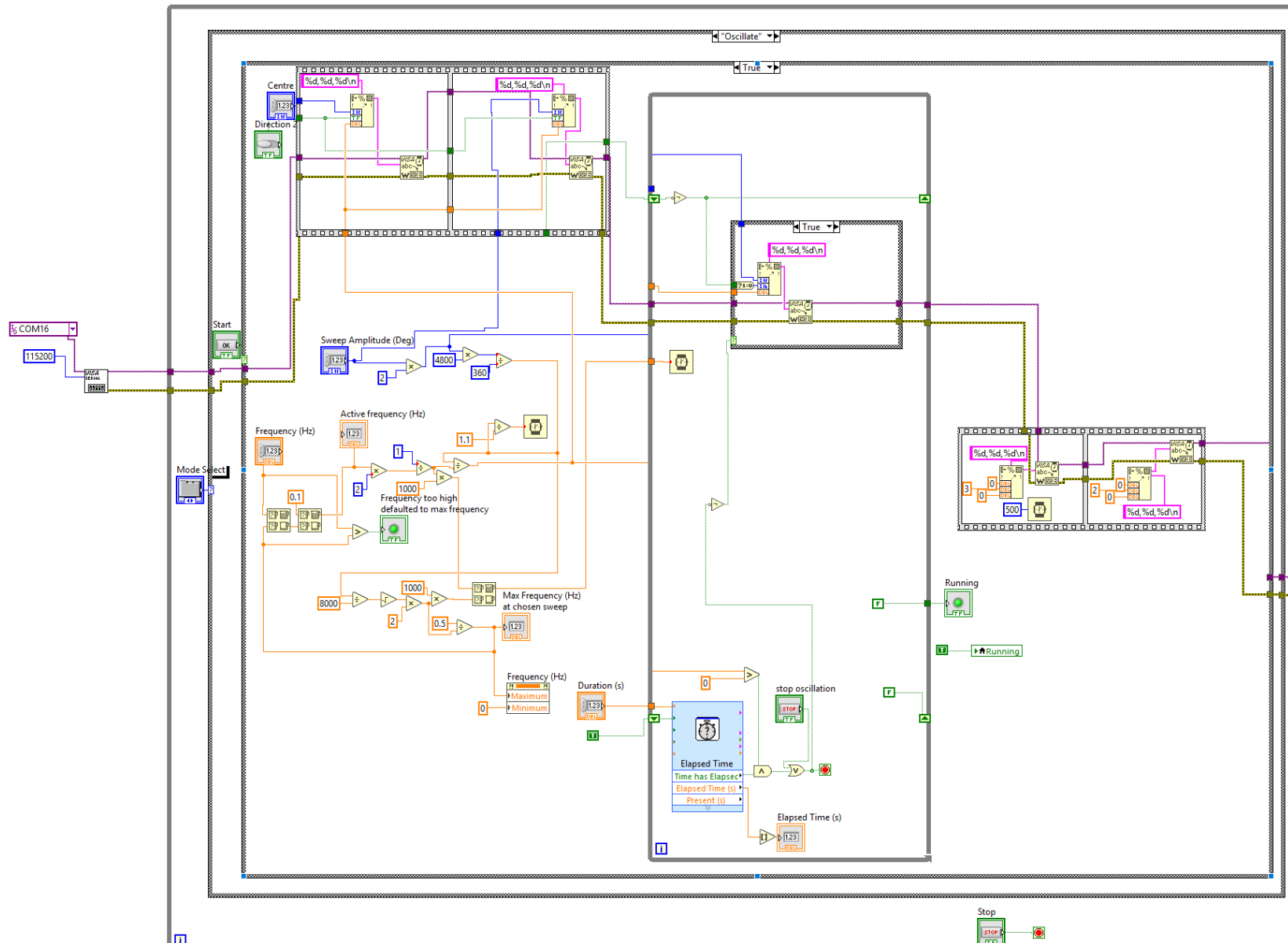


Figure D. 8: Oscillate mode block diagram

Receiver code

```
#include <Arduino.h>
#include <AccelStepper.h>
#include <esp_now.h>
#include <WiFi.h>

#define PUL_PIN 25
#define DIR_PIN 26

AccelStepper stepper(AccelStepper::DRIVER, PUL_PIN, DIR_PIN);

long homePosition = 0;
long pendingPulses = 0;
bool motorWasRunning = false;
volatile bool cancelMove = false;

typedef struct __attribute__((packed)) {
    int32_t angle;
    uint8_t direction; // 0=CCW, 1=CW, 2=HOME, 3=CLEARQ
    int32_t speed;
} MotorCommand;

#define QUEUE_SIZE 8
MotorCommand commandQueue[QUEUE_SIZE];
volatile int queueHead = 0;
volatile int queueTail = 0;

bool queueEmpty() { return queueHead == queueTail; }
bool queueFull() { return ((queueTail + 1) % QUEUE_SIZE) == queueHead; }

void flushQueue() {
    queueHead = 0;
    queueTail = 0;
}

void enqueue(MotorCommand cmd) {
    if (!queueFull()) {
        commandQueue[queueTail] = cmd;
        queueTail = (queueTail + 1) % QUEUE_SIZE;
    }
}

MotorCommand dequeue() {
    MotorCommand cmd = commandQueue[queueHead];
    queueHead = (queueHead + 1) % QUEUE_SIZE;
    return cmd;
}

void onReceive(const uint8_t *mac, const uint8_t *data, int len) {
    MotorCommand cmd;
```

```

memcpy(&cmd, data, sizeof(cmd));

if (cmd.direction == 3) {
    Serial.println("clear Q received");
    cancelMove = true;
    flushQueue();
    // stepper.motor(stop) CANNOT BE CALLED HERE DESTROYS IT FOR SOME REASON
    // gotta let it finish its move ig
    return;
}

enqueue(cmd);
}

void processCommand(MotorCommand cmd) {
    cancelMove = false;

    if (cmd.speed > 0) {
        stepper.setMaxSpeed(cmd.speed);
    }

    if (cmd.direction == 2) {
        stepper.stop();
        flushQueue();
        pendingPulses = 0;
        Serial.println("Homing...");
        Serial.print("Home position pulses: ");
        Serial.println(homePosition);
        long returnMove = -homePosition;
        if (returnMove > 2400) returnMove = returnMove - 4800;
        else if (returnMove < -2400) returnMove = returnMove + 4800;
        stepper.move(returnMove);
        homePosition = 0;
        return;
    }

    int pulses = (cmd.angle * 4800) / 360;
    if (cmd.direction == 1) {
        stepper.move(pulses);
        pendingPulses = pulses;
    } else {
        stepper.move(-pulses);
        pendingPulses = -pulses;
    }
}

void setup() {
    Serial.begin(115200);
    WiFi.mode(WIFI_STA);
    esp_now_init();
    esp_now_register_recv_cb(onReceive);
}

```

```

stepper.setMaxSpeed(16000);
stepper.setAcceleration(8000);
stepper.setPinsInverted(true, false, false);
stepper.setCurrentPosition(0);

Serial.println("Ready to receive;");
}

void loop() {
    stepper.run();

    if (motorWasRunning && !stepper.isRunning()) {
        if (pendingPulses != 0) {
            homePosition += pendingPulses;
            Serial.print("Home position: ");
            Serial.println(homePosition);
            pendingPulses = 0;
        }
    }
    motorWasRunning = stepper.isRunning();

    if (!queueEmpty() && !stepper.isRunning()) {
        MotorCommand cmd = dequeue();
        processCommand(cmd);
    }
}

```

Transmitter Code

```

#include <Arduino.h>
#include <esp_now.h>
#include <WiFi.h>

uint8_t receiverMAC[] = {0xE0, 0x8C, 0xFE, 0x59, 0x25, 0xB4}; // MAC address off ESP32
with sticker

typedef struct __attribute__((packed)) {
    int32_t angle;
    uint8_t direction; // 0=CCW, 1=CW, 2=HOME
    int32_t speed;
} MotorCommand;

MotorCommand command;

void onDataSent(const uint8_t *mac, esp_now_send_status_t status) {
    Serial.println(status == ESP_NOW_SEND_SUCCESS ? "Success" : "Failed");
}

void sendCommand(int angle, uint8_t direction, int speed = 0) {
    command.angle = angle;

```

```

    command.direction = direction;
    command.speed = speed;
    esp_now_send(receiverMAC, (uint8_t *)&command, sizeof(command));
}

void setup() {
    Serial.begin(115200);
    WiFi.mode(WIFI_STA);
    esp_now_init();
    esp_now_register_send_cb(onDataSent);

    esp_now_peer_info_t peerInfo = {};
    memcpy(peerInfo.peer_addr, receiverMAC, 6);
    peerInfo.channel = 0;
    peerInfo.encrypt = false;
    peerInfo.ifidx = WIFI_IF_STA;
    esp_now_add_peer(&peerInfo);

    Serial.println("Transmitter ready");
    Serial.println("Format: ANGLE,DIRECTION,SPEED");
    Serial.println("Example: 90,1 or 90,1,2400");
}

void loop() {
    if (Serial.available() > 0) {
        String input = Serial.readStringUntil('\n');
        input.trim();

        int commaIndex = input.indexOf(',');
        if (commaIndex != -1) {
            int angle = input.substring(0, commaIndex).toInt();
            int secondComma = input.indexOf(',', commaIndex + 1);
            uint8_t direction;
            int speed = 0;

            if (secondComma != -1) {
                direction = input.substring(commaIndex + 1, secondComma).toInt();
                speed = input.substring(secondComma + 1).toInt();
            } else {
                direction = input.substring(commaIndex + 1).toInt();
            }

            if (direction == 2) {
                Serial.println("Going home");
                sendCommand(0, 2, 0);
                return;
            }

            Serial.print("Sending: ");
            Serial.print(angle);
            Serial.print(" degrees, direction: ");

```

```
Serial.print(direction ? "clockwise" : "anticlockwise");  
Serial.print(", speed: ");  
Serial.println(speed);  
  
sendCommand(angle, direction, speed);  
} else {  
    Serial.println("Invalid format. Use: ANGLE,DIRECTION or ANGLE,DIRECTION,SPEED");  
}  
}  
}
```

11.4 Appendix E – Risk Assessment

Activity Being Assessed: Assembly, installation, operation and removal of an automated turntable system within the UWE large subsonic wind tunnel. The turntable is driven by [NAME OF MOTOR AND TORQUE VALUE], controlled by [DRIVER NAME and INPUT VOLTAGE FOR DRIVING AND LOGIC]. Motion commands are sent wirelessly through 5V ESP32 transmitter and receiver through 3.3V logic output. The assessment only covers the turntable system only – Other wind tunnel risks are addressed in separate risk assessment.	Assessed by: George Prior	Endorsed by:
Who might be harmed: UWE students and UWE technical and supervisory staff and site visitors How many exposed to risk: variable – max estimated: 8	Date of Assessment:	Review Date:

RISKS AND CONTROL MEASURES										
Hazards Identified	Existing Control Measures	S	L	Risk Level	Additional Control Measures	S	L	Risk Level	By whom and by when	Date Completed
Manual Handling during installation and removal (strains, sprains, crush, pinch point injuries)	• Minimum two person lift for turntable assembly and removal • Clear working space maintained around turntable and test section at all time • Trolley used to transport heavy components to the tunnel • Steel-capped footwear worn throughout	3	2	6	• One person designated to coordinate the lift • Pinch points between turntable disc and tunnel test section floor identified before installation begins • Turntable system bolted to tunnel floor before system is powered on	3	1	3		
Rotating machinery – drive belt, pulleys, shaft, motor and disc (entanglement, nip-point contact, crush injuries)	• Belt, pulleys and motor assembly enclosed or guarded • Drive system powered off before any access to drive components • Loose clothing, jewellery and hair tied back when near turntable • Drive system only powered back on when all personnel are clear of the mechanism	4	2	8	• LOTO applied before any maintenance on drive system • Emergency stop procedure defined and communicated to all users before use • Physical guard fitted over pulley and belt mechanism if not fully enclosed by turntable housing	4	1	4		

Electrical hazards – motor driver, ESP32 controllers, DAQ and motor wiring (electric shock, burns from incorrect wiring or short)	• All wiring and connections made and inspected before system is initially powered on • Motor driver powered at rated voltage only, with mains supply fused suitably • Motor phase and encoder wires labelled and connected per driver manual	4	2	8	• Power down before any wiring changes • Initial powering on supervised by UWE technical staff	4	1	4		
Unexpected turntable actuation (injury from sudden rotation of disc or model mounted, or from drive mechanism below)	• Wireless link only active when intentionally control from DAQ • Driver enable (ENA) pin used to prevent motion unless testing is ready to begin • All personnel clear of the turntable before motion command is sent	4	2	8	• Start sequence: safety check, all clear confirmed verbally and visually, motion enabled • Emergency stop location and procedure known to all present • Turntable not actuated while personnel are near the test section	4	1	4		
Slips, trips and falls from control system wiring (Injury from DAQ and power supply wiring on the floor)	• All cables routed and secured along walls and floors secured • Walkways around turntable and control equipment always kept clear • Adequate lighting maintained in test areas	2	3	6	• Checks carried out at start and end of session ensuring walkways are kept clear • Temporary cable runs crossing walkways marked clearly	2	1	2		

11.5 Appendix F – Testing Tables

Test	Angle (deg)	Repetitions	Result
Cumulative error test	1	360	Returned to home, within 0.5° tolerance
Cumulative error test	5	72	Returned to home, within 0.5° tolerance
Cumulative error test	10	36	Returned to home, within 0.5° tolerance
Cumulative error test	20	18	Returned to home, within 0.5° tolerance
Homing test	various	50 random moves repeated, 5 times	Successfully returned home within 0.5° buffer every time

Table F. 1: Manual angle control and homing testing table

Sweep amplitude (deg)	Centre angle (deg)	Frequency (Hz)	Result
30	0	0.25	Consistent bidirectional movement within 0.25 second frequency buffer
30	0	0.5	Consistent bidirectional movement within 0.25 second frequency buffer
15	45	0.25	Consistent bidirectional movement within 0.25 second frequency buffer
10	90	1	Consistent bidirectional movement

Table F. 2: Oscillation control testing table